

SMART INDIA HACKATHON 2026 · INTERNAL REVIEW

ANNA DRISHTI

TECHNICAL HANDBOOK & JUDGE-DEFENCE MANUAL

Everything behind the fourteen slides — the grain science, the transport physics, the sensing, the electronics, the mathematics, the validation, and every number audited against its source. Written so that after studying it you can explain this project to a farmer and defend it to a professor.

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Source deck ANNA DRISHTI, 14 slides · audited 25 August 2026

How to use this handbook

Read this page first. It explains the evidence-classification system, without which the rest of the document can be misread — and misreading it in front of a judge is worse than not reading it at all.

The evidence classification system

Every technical claim in this handbook carries a tag. The tag tells you **how much weight the claim can bear under questioning**. This matters more than whether the claim sounds impressive: a judge who catches you stating an assumption as a fact will discount everything else you say.

TAG	WHAT IT MEANS	HOW TO SPEAK IT
ESTABLISHED	Textbook physics, chemistry or biology. Not in dispute.	State it plainly. “Grain is an insulator.”
LITERATURE	Supported by a specific published source we can name.	Name the source. “A 2023 study in Scientific Reports found...”
ENGINEERING EST.	A defensible calculation from accepted constants, but our arithmetic, not a published result.	Say it is modelled. “Our model gives roughly...”
ASSUMPTION	A value we chose to make the model runnable. Reasonable, but chosen.	Own it. “We assumed X, and here is why that is conservative.”
HYPOTHESIS	What we expect to happen but have not shown.	Say “we expect”, never “we found”.
NEEDS VALIDATION	Cannot be defended today. An honest gap.	Volunteer it before you are asked.

THE SINGLE RULE THAT PROTECTS YOU

If a judge asks “**how do you know?**”, your answer must identify which of the six categories above the claim sits in. “That is an assumption we made, and here is why we chose it” is a *strong* answer. “It is in our model” is not an answer at all — it describes where the number lives, not why it is right.

What this handbook is built from

SOURCE	WHAT WAS TAKEN FROM IT
The 14-slide ANNA DRISHTI deck	Every claim, number, table and diagram. Text extracted directly from the file, not retyped.
The simulation source (<code>index.html</code>)	Every constant, equation and threshold — and their exact provenance, because the model is where most of the deck’s numbers were computed.
The submitted SIH proposal	The twelve-item reference list and the institutional framing.
Independent verification	Literature checks on the load-bearing claims, carried out for this handbook. Results in Part D §15 and flagged inline wherever they change what you should say.

NOTHING HERE IS INVENTED

No citation, dataset, experimental result, cost or competitor capability in this handbook was fabricated. Where something is unknown it says **NEEDS VALIDATION** and stops. That restraint is the point of the document — a handbook that makes the project sound better than it is would fail you at exactly the moment it mattered.

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The audit

Nine findings from checking every claim in the deck against the simulation source and the literature. Read this section even if you read nothing else. Four of these change what you should say out loud.

0.1 Findings that change your wording

FINDING 1 — “PHYSICALLY CANNOT” IS AN OVERCLAIM (SLIDE 5)

The deck says: “0.82 — GROWTH FLOOR. Below this *A. flavus* physically cannot pull water out of the grain. Not ‘slow’ — cannot.”

The problem: reported minimum water activity for *A. flavus* growth **varies by strain, substrate and temperature**. Published values range from roughly 0.82 up to 0.91 depending on the study — one study on sorghum reported a minimum of 0.91 for mycelial growth, while classical work on corn proposed values around 0.83–0.84 for toxin production. 0.82 is the *conservative lower end of a reported range*, not a physical constant.

Why it matters: a mycology-literate judge knows the spread. The word “cannot” invites a correction that makes every other number you state look equally overconfident.

Say this instead: “0.82 is the conservative end of the published range for *A. flavus* growth — reported minima vary with strain and substrate, some studies report higher. We deliberately took the lowest credible value, because for a warning system you want to alert early rather than late.” **That answer is stronger**, because it shows you know the literature has spread and chose your side of it deliberately.

FINDING 2 — THE 1.6 M FIGURE NEEDS ONE QUALIFYING WORD (SLIDE 7)

The deck says: “A spear reaches forty centimetres. Our hotspot is at one point six metres.”

What the model actually contains: the demo pocket sits at grid cell (24, 16) in a 4.0 × 2.5 m stack. Its distances to the four faces are:

FACE	DISTANCE	SPEAR-ACCESSIBLE?
Top	1.6 m	Yes — this is the 1.6 m in the deck
Right side	1.5 m	Yes
Left side	2.4 m	Yes
Bottom	0.8 m	No — that is the floor

The risk: if a judge mentally pictures probing from the nearest surface and says “isn’t it only 0.8 m from the bottom?”, an unprepared answer looks like you cherry-picked the flattering number.

Say this: “1.6 m from the top face, 1.5 m from the nearest side. The floor side is 0.8 m but a stack sits on dunnage — you cannot spear upward through the floor. So the nearest *reachable* face is about 1.5 m, which is nearly four times a spear’s reach.” **Bonus point available:** our simulation is actually *generous* to the baseline here — it lets the inspector sample inward from all four edges including the floor, which is physically impossible. We handicapped ourselves.

FINDING 3 — “~15×” UNDERSTATES YOUR OWN MODEL (SLIDE 3)

The deck says: “SO GAS LEADS BY ~15×”.

The model gives 16.7×, both as a ratio of arrival times ($15.43 \text{ d} \div 0.926 \text{ d}$) and as a ratio of diffusivities ($2.0 \times 10^{-6} \div 1.2 \times 10^{-7}$). These agree exactly because arrival time scales as L^2/D , so the length cancels.

This is a harmless rounding down — but know the exact figure, because “16.7, we round it down to about 15 when we speak” sounds like command of your own model, while being unable to reproduce your own headline ratio does not.

FINDING 4 — THE SHT31 COST LINE IS 4× WHAT YOU ACTUALLY PAID (SLIDE 12)

The deck says: “3× SHT31 ₹3,000” (i.e. ₹1,000 each).

Your receipt says: 3 × SHT3X at ₹250 each = **₹750**.

The slide’s total of ₹9,000–11,000 is still internally consistent (₹10,450 by its own line items), so nothing collapses. But if you show a receipt or a judge prices the parts, the discrepancy looks careless. **Recomputed with your real price the total is ₹8,200** — which is a *better* number. Either correct the slide or be ready to say “that line is conservative; we actually sourced them at ₹250 each.”

0.2 Findings that confirm the deck is right

CLAIM	VERIFICATION	VERDICT
Phosphine resistance 2.11–70.94-fold (Slide 7)	Verified against the source. <i>Scientific Reports</i> vol. 13, article 16497 (2023), “Current status of phosphine resistance in Indian field populations of <i>Tribolium castaneum</i> ...” — LC ₅₀ values 0.038–1.277 mg L ⁻¹ , 2.11 to 70.94-fold resistance. The deck quotes this exactly, including the volume and article number.	EXACT
Detection day 0.8 vs day 42 (Slide 8)	Re-ran the simulation at hourly granularity: Anna-Drishti flags at day 0.79 , conventional inspection at day 42 , actual mould onset day 9.7. The deck rounds 0.79 to 0.8.	EXACT
“About 15 % moisture” (Slide 5)	Solved the isotherm for the $a_w = 0.82$ crossing: 14.66 % wb at 30 °C , 14.88 % at 25 °C. “About 15 %” is fair.	FAIR
Water-activity table (Slide 5)	All five rows reproduce the model exactly: 12 %/30 °C→0.657, 14/25→0.774, 16/30→0.875, 18/25→0.926, 20/30→0.963. The time-to-mould column also matches (0.95→6.7 d, 0.90→10.8, 0.87→17.4, 0.85→28.9, 0.83→86.8).	EXACT
Sigma separation 1.5 / 3.0 / 4.2 (Slide 10)	Matches the measured values: clean grain peaks at 1.5 σ over ten simulated days, alert threshold 3.0 σ , real hotspot reaches 4.2 σ .	EXACT
“1.2 m is furthest from any face” (Slide 12)	In a 2.5 m stack, the 1.2 m port is 1.2 m from the top and 1.3 m from the bottom — minimum 1.2 m. The 0.5 m and 2.0 m ports are both 0.5 m from a face. Correct.	CORRECT
Cost total (Slide 12)	Line items sum to ₹10,450, inside the stated ₹9,000–11,000 band and inside the ₹19,800 proposal budget. Arithmetic is sound (see Finding 4 on the SHT31 line).	CONSISTENT

0.3 One internal inconsistency worth tidying

SLIDE 9 SAYS ± 2 % RH; SLIDE 10 SAYS ± 1.0 % RH

Both are defensible but they are different quantities, and saying them in the same breath sounds like a mistake:

- **± 2 % RH** (slide 9) is the SHT31 *datasheet accuracy specification* — the manufacturer’s stated tolerance band.
- **± 1.0 % RH** (slide 10) is the *one-sigma noise* we inject into the simulation’s virtual sensors.

These are consistent if you read the datasheet band as roughly two sigma, which is the normal convention. **If asked:** “Two percent is the datasheet accuracy band; one percent is the standard deviation we simulate, which is the same thing expressed as one sigma.”

0.4 The gap the deck already admits — and should keep admitting

Slide 11 states plainly that you have **not** measured CO₂ and have **not** observed spoilage. Slide 12 leaves the coverage result as an explicit placeholder (“detect N % of hotspots within X days”).

DO NOT REMOVE THESE

Volunteering a limitation before a judge finds it is the highest-value move available to you, and slide 11 is the strongest slide in the deck for exactly that reason. The placeholder on slide 12 is the one exception — **either fill it with a real number before the review or delete the line**. A visible “to be inserted” on a presented slide reads as unfinished work, whereas an honest “we have not run that analysis yet” spoken aloud reads as candour. Same fact, opposite impression.

The project from zero

Two explanations of the same thing. The first assumes nothing. The second assumes you have read the first. You need both, because judges ask for both — often in the same minute.

1 Anna-Drishti explained to a non-specialist

THE 30-SECOND VERSION, IN PLAIN WORDS

“Grain in an Indian warehouse is stored in sacks, stacked several metres deep. If damp gets into the middle of that stack — a roof leak, a wet sack, condensation — the grain there starts to rot from the inside. Nobody can see it, because it is buried under a metre and a half of other sacks.

Now, rotting grain **breathes**. The grain and the mould growing on it consume sugar and oxygen and give off carbon dioxide, water and heat — the same thing a compost heap does. That carbon dioxide seeps out through the air gaps between grains fairly quickly. The heat has to crawl through the grain itself, and grain is a very good insulator, so the heat takes far longer to reach anywhere you could measure it.

So we push thin perforated tubes into the stack, sample the air deep inside, and watch the carbon dioxide. It tells us something is going wrong roughly two weeks before a temperature probe would notice. And instead of just showing a number, we tell the warehouse manager which stack, which depth, what to do, and how many days they have.”

The four things a non-specialist needs to accept

CLAIM	EVERYDAY VERSION
Grain is alive	A seed is a living thing waiting to sprout. It breathes slowly. So does anything growing on it.
Breathing makes gas, heat and water	Same as a compost heap: warm in the middle, and giving off gas.
Gas travels faster than heat here	Gas slips through the gaps between grains. Heat has to soak through the grains themselves, and grain insulates — that is why a sack of rice keeps its temperature for hours.
Once the toxin forms it cannot be undone	You can kill the mould. You cannot un-poison the grain. So the only useful moment is <i>before</i> .

THE ONE ANALOGY THAT RELIABLY LANDS

A compost heap. Everyone has seen one steam in the cold and smelled it from a distance. The smell reaches you across the garden long before you could feel the warmth by touching it. A spoiling pocket inside a grain stack is the same event — smaller, slower, and buried. We are standing where the smell arrives, not where the warmth does.

2 Anna-Drishti explained technically

Anna-Drishti is a **depth-resolved interstitial gas and hygrometry sensing system** coupled to a **predictive interpretation layer**, deployed in bagged-grain storage.

Three perforated probe lances are inserted between bag courses across a stack face. Each lance carries sampling ports at three depths, giving nine sampling points. A single NDIR CO₂ analyser is sequenced across all nine ports through a manifold; humidity and temperature are measured locally at each port with digital sensors.

The interpretation layer performs four conversions:

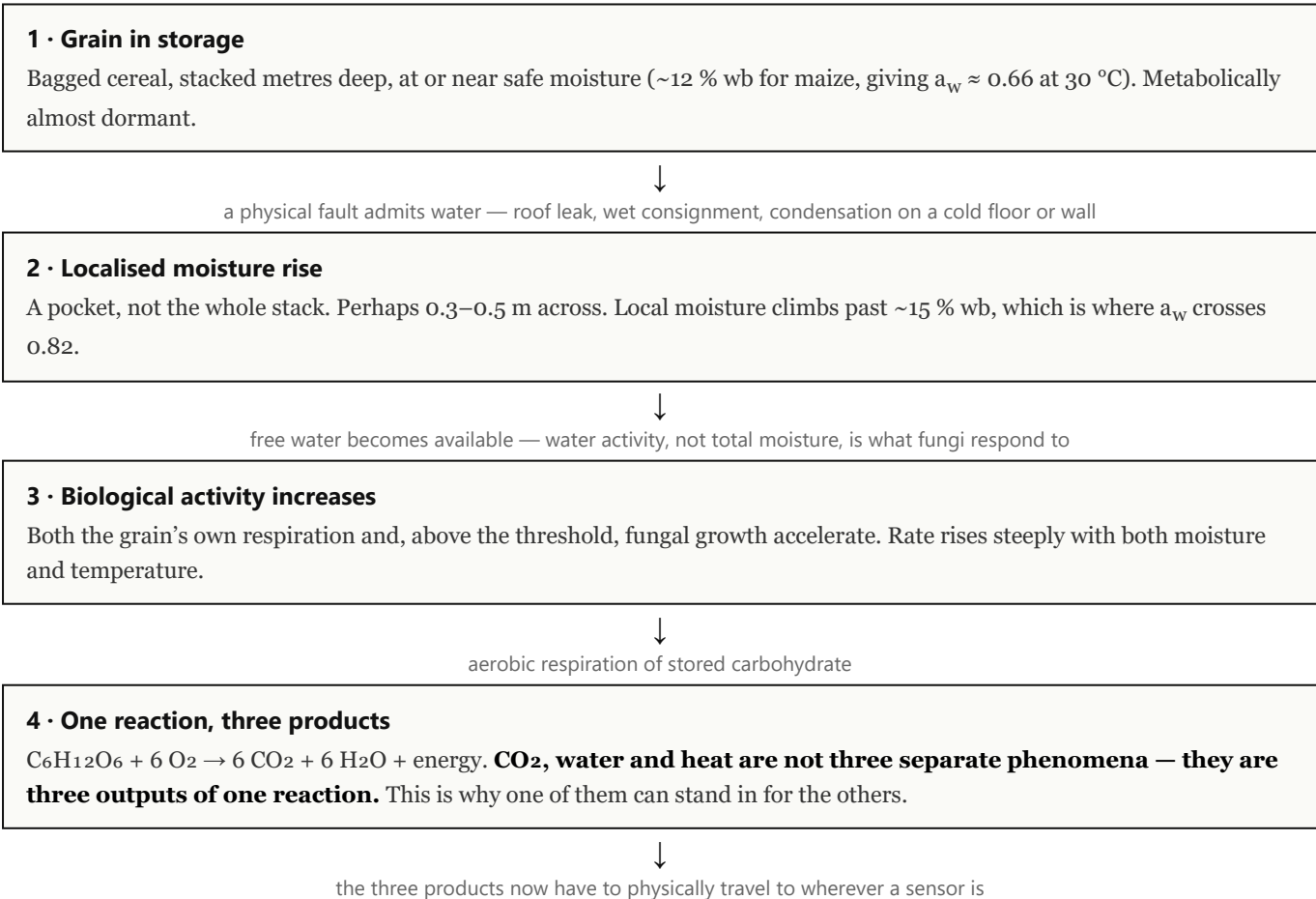
STEP	CONVERSION	METHOD
1	Relative humidity → water activity	At equilibrium, interstitial RH is the grain’s water activity. No model needed — it is a definition.
2	a _w + temperature → days to mould onset	Cardinal-value predictive mycology model for <i>A. flavus</i>
3	CO ₂ → respiration anomaly	Depth-matched differential comparison across lances, in units of sigma
4	All three → ranked action	Composite risk score → ranked ports → action card with a deadline

THE DISTINCTION THAT DEFINES THE PROJECT

The output is not a measurement. It is a **dated, evidence-carrying decision**: which stack, which depth, which action, by when, and on what basis. A system that reports 4,200 ppm has told a warehouse manager nothing actionable. A system that says “*Lance 2 at 2 m, water activity 0.91, modelled mould onset in six days, act by the 30th*” has.

3 The full causal chain — every arrow explained

This is the spine of the project. Each box is a physical state; each arrow is a mechanism. If you can narrate this chain unprompted, you can answer most questions by locating them on it.



5 • Three signals, three very different speeds

CO₂ moves as a gas through interstitial air. Heat conducts through kernels and trapped air. Moisture migrates slowest of all. Over 0.4 m our model gives roughly **0.9 days, 15.4 days and 123.5 days** respectively.



the gas reaches a sampling port first — this is the entire thesis of the project

6 • Detection at the port

Nine ports sampled by one shared analyser. A port is compared against the other lances *at its own depth*, in sigma. Alert at 3.0 σ sustained over three consecutive readings.



ESP32 acquires, timestamps, CRC-checks and transmits

7 • Interpretation

RH becomes water activity. a_w and temperature become days-to-onset. CO₂ becomes an anomaly score. Three weak signals become one confident estimate.



composite risk score, zones ranked by urgency

8 • Decision, not alarm

An action card: stack, depth, recommended action, deadline, and the evidence behind it.



a human acts — aerate, restack, or targeted fumigation

9 • Verification — the loop closes

Post-intervention CO₂ decay confirms the treatment worked. If the anomaly does not fall, the system escalates rather than assuming success. **Current practice never checks at all.**

IF A JUDGE STOPS YOU ANYWHERE ON THIS CHAIN

Name the box, name the mechanism on the arrow into it, and name the evidence. For example, at box 5: "That arrow is diffusive transport. The mechanism is that CO₂ moves through the connected air space between kernels while heat has to conduct through the kernels themselves. Our diffusivity ratio is about 17 to 1, which is an engineering estimate from published constants, not a measurement we made."

The problem

The deck spends slides 2 and 7 here. This part gives you the depth behind them — enough to keep answering after the slide has moved on.

4 Indian grain storage and the bag stack

India’s public grain storage is dominated by **bagged stacks in godowns**, not bulk silos. Grain sits in jute or HDPE sacks, stacked on wooden dunnage to keep it off the floor, in regular rectangular stacks several metres deep and several metres tall.

WHY THIS GEOMETRY CHANGES EVERYTHING

PROPERTY	BULK SILO	INDIAN BAG STACK
Access to interior	Auger, gravity, or fixed cables	Physically obstructed by bags. Probing damages sacks.
Air paths	Uniform interstitial voids	Voids between kernels <i>and</i> between bags — more complex, less uniform
Instrumentation	Permanent, per-vessel	Stacks are built, moved and dismantled. Permanent fitting is uneconomic.
Commercial products	Mature market	Designed for the silo case, priced for it

THE GAP, STATED PRECISELY

There is no affordable, non-destructive, interior-resolved, predictive monitoring system designed for the **bag-stack storage geometry** that dominates Indian public and cooperative warehousing. [LITERATURE](#) for the components; the gap statement itself is our synthesis.

THE INSTITUTIONAL CONTEXT — WHY THE PROBLEM IS GROWING

Under the **World’s Largest Grain Storage Plan in the Cooperative Sector**, godown capacity is being built at Primary Agricultural Credit Society level, with 500+ PACS godowns identified for construction. [LITERATURE](#) These facilities will have **neither instrumentation nor trained storage scientists on site**. Capacity is being distributed faster than the expertise to monitor it.

5 How a wet pocket forms, and why it accelerates

FOUR COMMON ORIGINS

1. **Roof leaks during monsoon** — water enters at the top of a stack.
2. **A wet consignment** — grain loaded slightly above safe moisture into one corner.
3. **Condensation** — day–night temperature swings drive moisture to cold surfaces: outer walls, the floor, the shaded side.
4. **Moisture migration** — temperature gradients drive slow convection currents that redistribute moisture within the bulk over weeks.

WHY IT DOES NOT STAY SMALL — THE FEEDBACK LOOP

Higher moisture -> faster respiration
faster respiration -> releases HEAT and WATER
more heat + more water -> higher moisture and higher temperature
-> faster respiration (loop closes)

This is a **positive feedback**, which is why spoilage in grain is not linear. Small and slow for a long time, then abruptly not. In severe cases the loop runs to visible self-heating. **ESTABLISHED**

THE PRESENTATIONAL CONSEQUENCE
This is *why* early detection is worth so much more than accurate late detection. On a linear process, detecting at 50 % of the way through saves you half. On an accelerating process, detecting early saves you almost everything — and detecting late saves you almost nothing.

6 Why inspection physically cannot reach it

Slide 7 makes the crucial framing point: **this is not a criticism of inspectors**. It is a statement about geometry. Never let this come across as “inspection is bad” — that is both unfair and easy to attack.

HOW INSPECTION ACTUALLY WORKS

An inspector walks the stack looking for staining, visible mould and insect exit holes, then pushes a slotted hollow spear into accessible bags and examines the retained sample. Careful, skilled work — and limited in exactly three ways:

#	LIMITATION	WHY IT IS STRUCTURAL, NOT FIXABLE BY EFFORT
1	It only reaches the outside	A bag spear reaches roughly 0.4 m. A godown stack is metres deep. The centre is not reachable by any amount of diligence. ASSUMPTION on the exact figure — see §15.
2	Problems are not spread evenly	FAO guidance is explicit that insects concentrate in dust pockets and zones of heating. A clean sample from one spot says nothing about a pocket two metres away. LITERATURE
3	Early stages are invisible	Eggs and larvae develop concealed inside kernels. FAO notes that absence of live adults in a sample does not indicate absence of infestation. LITERATURE

THE LINE FROM SLIDE 7, AND HOW TO DELIVER IT
“A spear reaches forty centimetres. Our hotspot is at one and a half to one point six metres from the nearest reachable face. No amount of diligence gets a spear to that location. This is not a criticism of inspection — it is precisely the reason an interior probe needs to exist.”
Pause after “that location.” It is the strongest sentence in the deck and it needs a beat.

7 The mycotoxin stakes

Quantity loss is the obvious cost and it is not the important one. Storage conditions determine **mycotoxin risk**, and mycotoxin risk is qualitatively different because it is irreversible.

FACT	CONSEQUENCE FOR THE DESIGN
<i>A. flavus</i> requires a minimum water activity to grow — conservatively 0.82 LITERATURE	There is a specific measurable threshold . This is what makes prediction possible at all — without a threshold there is nothing to forecast the crossing of.
Toxin production needs a narrower window than growth LITERATURE	Two thresholds must be modelled, not one. Growth first, then toxin.
Aflatoxin is not destroyed by drying , and is heat-stable ESTABLISHED	Post-hoc detection is useless for intervention. The problem is inherently predictive.
Aflatoxin B1 is a liver carcinogen ESTABLISHED	Food-safety and export-trade dimensions on top of the economic one.

THE BOUNDARY YOU MUST STATE BEFORE A JUDGE FINDS IT

“We do not measure toxin. We predict the conditions that create it, and we say so in every output the system produces.”

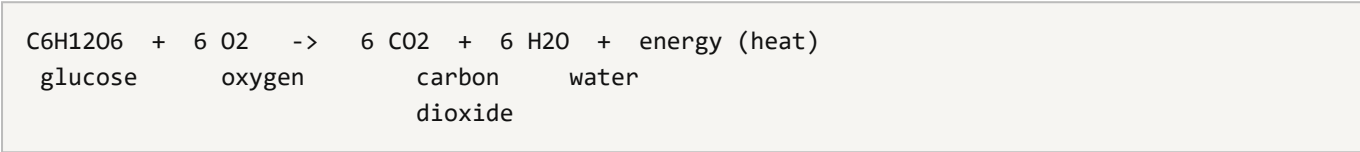
Confirming toxin requires a laboratory assay. A system claiming to read aflatoxin off a humidity sensor would be scientifically indefensible, and any competent judge would know it immediately. Slide 6 exists to say this out loud — **it is the most credibility-building slide in the deck**. Do not skip it to save time.

Grain science

The biology and physical chemistry underneath slides 3 and 5. This is Krittika’s territory, but every member should be able to give the first two paragraphs of each section.

8 Respiration — the single reaction everything rests on

THE REACTION



Both the grain itself (a living seed) and any fungi growing on it run this reaction. It is ordinary aerobic respiration.

ESTABLISHED

WHY THIS SINGLE EQUATION CARRIES THE WHOLE PROJECT

CO₂, water and heat are **not three independent phenomena that happen to occur together**. They are three products of one reaction, in fixed stoichiometric proportion. That is what licenses using one of them as a proxy for the others — and it is the answer to “why should CO₂ tell you anything about heat or spoilage?”

THE FIXED CHEMISTRY — NUMBERS THAT ARE NOT TUNABLE

From the molar masses (180 g glucose yields 264 g CO₂):

QUANTITY	VALUE	MEANING
CO ₂ per dry matter lost	1.467 g/g	264 ÷ 180. Pure stoichiometry.
1 % dry-matter loss	14.67 g CO ₂ /kg	Lets us convert accumulated CO ₂ into grain actually lost
Heat per gram CO ₂	10,670 J	2,816 kJ/mol glucose ÷ 264 g CO ₂

ESTABLISHED — these follow from the reaction equation. If a judge challenges them, you can derive them on paper. That is a good moment to have available.

HOW FAST DOES GRAIN RESPIRE?

Respiration rate rises steeply with both moisture and temperature. Our model uses:

$$R = R_0 \times \exp(k \times (M - 14)) \times Q_{10}^{((T - 25)/10)}$$

$R_0 = 0.5 \text{ mg CO}_2 / \text{kg dry matter} / \text{day}$ at 14% wb, 25 C

$k = 0.7$ per moisture point

$Q_{10} = 2.5$ (rate multiplies by 2.5 per 10 C rise)

MOISTURE (WB)	TEMP	RESPIRATION MG CO ₂ /KG/DAY	INTERPRETATION
12 %	30 °C	0.19	Safe storage. Nearly dormant.
14 %	25 °C	0.5	Reference point

MOISTURE (WB)	TEMP	RESPIRATION MG CO ₂ /KG/DAY	INTERPRETATION
16 %	30 °C	3.2	17× the safe rate
18 %	30 °C	13	68×
20 %	30 °C	52.7	277× — the runaway regime

ENGINEERING EST. The functional *form* (exponential in moisture, Q₁₀ in temperature) is standard. The specific coefficients are ours, chosen so the model reproduces published safe-storage timelines. **They need Krittika’s sign-off and are listed on her verification sheet.**

THE NUMBER TO REMEMBER

“Going from 12 % to 20 % moisture increases respiration by more than two orders of magnitude. That is why a small wet pocket produces a gas signal far out of proportion to its size — it is doing almost all of the breathing in the stack.”

9 Moisture, humidity and water activity

Slide 5’s headline is “**Moisture is the total. a_w is the spendable part.**” This is the single most important concept in the biology, and the one most likely to be probed.

THREE QUANTITIES PEOPLE CONFUSE

QUANTITY	DEFINITION	WHAT IT TELLS YOU
Moisture content (% wet basis)	Mass of water ÷ total mass, as a percentage. What a moisture meter reads.	How much water is present <i>in total</i> — including water chemically bound to starch and protein that no organism can use.
Relative humidity (% RH)	Water vapour pressure in air ÷ saturation vapour pressure, as a percentage.	The state of the <i>air</i> , not the grain — until they reach equilibrium.
Water activity (a _w , 0–1)	The <i>free</i> fraction of water — chemically available, not bound.	What organisms actually respond to. This is the biologically meaningful quantity.

THE EQUIVALENCE THE WHOLE SENSING STRATEGY DEPENDS ON

At equilibrium, the relative humidity of the air between the kernels IS the water activity of the grain. $a_w = RH \div 100$.

This is not a model or an approximation we invented — it is what water activity means. It is why a ₹250 humidity sensor can measure the variable that governs fungal growth, and why we do not need a laboratory water-activity meter in the field. **ESTABLISHED**

WHY THE SAME MOISTURE BEHAVES DIFFERENTLY AT DIFFERENT TEMPERATURES

Temperature changes how tightly water is held by the starch and protein matrix. Warmer grain releases a larger free fraction at the same total moisture. This is why slide 5’s table has both a moisture column *and* a temperature column — and why quoting a moisture percentage alone is scientifically incomplete.

THE SORPTION ISOTHERM — CONVERTING BETWEEN THEM

We use the **modified Chung–Pfof equation** with published ASAE D245 coefficients for shelled corn:

LITERATURE

$$a_w = \exp\left(-\left(\frac{A}{T + C}\right) \times \exp(-B \times M_{db})\right)$$

A = 620.7, B = 0.2118, C = 52.15 (shelled corn, ASAE D245)

T = temperature in C

M_db = moisture, dry basis %

Slide 5's table is this equation evaluated at five points. All five reproduce exactly (§0.2). **Why this matters rhetorically:** we did not invent a conversion. We used a published standard with published coefficients for the specific commodity in our jars.

THE TWO THRESHOLDS

A _w	WHAT HAPPENS	CONFIDENCE
0.82	Growth floor. Below this, <i>A. flavus</i> growth is not supported in the literature we rely on. Reported minima vary with strain and substrate; 0.82 is the conservative end of the published range.	LITERATURE see Finding 1
0.85	Toxin line. Aflatoxin production requires conditions wetter than growth alone.	LITERATURE

In practical terms, at 30 °C the $a_w = 0.82$ crossing occurs at **14.66 % moisture wet basis**. Slide 5's "about 15 %" is fair. **ENGINEERING EST.**

10 What actually happens inside a wet pocket

Putting Parts B and C together — the full physical, biological and chemical sequence:

PHASE	TIMESCALE	WHAT IS HAPPENING
0	Instant	Water arrives. Local moisture rises. Nothing biological yet.
1	Hours	Water redistributes into kernels and the interstitial air equilibrates. Local a_w rises past 0.82. <i>The clock starts here.</i>
2	Hours–days	Respiration accelerates — grain first, then fungal as spores germinate. CO ₂ production climbs steeply. This is what we detect.
3	~1 day over 0.4 m	CO₂ reaches a sampling port. Detectable as a differential anomaly against other ports at the same depth.
4	Days	Fungal growth establishes. At a_w 0.92, 30 °C our model gives visible mould in ~8.7 days.
5	~15 days over 0.4 m	Heat finally reaches a temperature probe — roughly two weeks after the gas did. A temperature-based system notices around here.
6	Days–weeks	Dry matter loss accumulates; toxin risk develops if a_w and temperature are both in the toxin window. Now irreversible.
7	Weeks	Externally visible — staining, musty odour, warm bags. Conventional inspection notices around here. Our model puts this at day 42.

THE ONE-SENTENCE VERSION OF THIS ENTIRE TABLE

"The same event is detectable as gas on day one, as heat around day fifteen, and by smell around day forty. We chose to stand where it arrives first."

The physics: spoilage breathes

Slide 3 is the intellectual centre of the project. This part explains why it is true, how the numbers were produced, and exactly how much weight each one can carry.

11 Why gas arrives first

All three signals — CO₂, heat, moisture — leave the same pocket at the same moment. They arrive at a sensor at wildly different times because they travel by **different physical mechanisms through different media**.

SIGNAL	TRAVELS BY	THROUGH WHAT	WHY FAST OR SLOW
CO ₂	Molecular diffusion of a gas	The connected air space between kernels	Fast. Gas molecules move freely through open, connected voids. Roughly 40 % of a grain bed by volume is air.
Heat	Thermal conduction	The kernels themselves , plus trapped air	Slow. Grain is a good thermal insulator, and the trapped air pockets that help gas move actually <i>hinder</i> conduction.
Moisture	Vapour diffusion + sorption	Air, but repeatedly absorbed by grain	Slowest. Every kernel it passes acts as a sponge, absorbing and re-releasing it. The grain buffers moisture heavily.

THE ELEGANT PART — AND WORTH SAYING OUT LOUD

The **same property** that makes grain a good insulator (trapped air between kernels) is what makes it a good conductor of gas. Porosity helps CO₂ and hinders heat. They are not two unrelated facts — they are the same fact seen from two sides. **ESTABLISHED**

12 Fick’s law — how gas moves

INTUITIVELY

Put a drop of ink in still water. It spreads out, from where there is a lot to where there is little, and it never spontaneously re-concentrates. The steeper the difference in concentration, the faster it spreads. That is diffusion, and Fick’s law is just that sentence written in symbols.

TECHNICALLY

Fick's First Law

$$J = -D \frac{dC}{dx}$$

J

= flux

(how much gas crosses unit area per second)

D

= diffusivity

(m²/s - how easily the medium lets it through)

dC/dx

= concentration gradient

minus

= flow goes DOWN the gradient, high to low

For how a concentration *evolves* over time, Fick’s second law applies. The practical consequence we use is the characteristic diffusion time:

$$t = L^2 / D$$

t = time to travel distance L

Note the SQUARE. Doubling the distance quadruples the time.

THE ONE INSIGHT TO TAKE FROM THIS EQUATION

Because distance is **squared**, diffusion is fast over short distances and punishingly slow over long ones. This is exactly why a probe placed *inside* the stack works and a sensor on the outside does not — not because the outside sensor is worse, but because it is asking diffusion to do something it is bad at.

13 Fourier's law — how heat moves

INTUITIVELY

Hold a metal spoon in hot tea and the handle warms in seconds. Hold a wooden spoon and it barely warms at all. Same heat, same geometry — different material. Grain behaves like the wooden spoon.

TECHNICALLY

Fourier's Law $q = -k \, dT/dx$

q = heat flux

k = thermal conductivity (W/m.K)

dT/dx = temperature gradient

Thermal diffusivity $\alpha = k / (\rho \times c_p)$

ρ = density, c_p = specific heat capacity

α has the SAME units as D (m^2/s) - which is what lets us compare gas transport and heat transport directly.

WHY THE COMPARISON IN SLIDE 3 IS LEGITIMATE

A judge may reasonably ask whether comparing a gas diffusivity with a thermal diffusivity is apples-to-oranges. It is not. **Both have units of m^2/s , and both obey a $t = L^2/D$ scaling law.** They are mathematically the same kind of quantity describing the same kind of spreading process. That is precisely why the ratio is meaningful. **ESTABLISHED**

14 The three timescales compared

Our model uses these three transport coefficients:

QUANTITY	VALUE (M^2/S)	TIME TO CROSS 0.4 M	EVIDENCE CLASS
CO ₂ effective diffusivity in interstitial air	2.0×10^{-6}	0.93 days	ENG. EST.
Thermal diffusivity of bulk grain	1.2×10^{-7}	15.43 days	ENG. EST.
Effective moisture migration coefficient	1.5×10^{-8}	123.46 days	ASSUMPTION

Worked example - the CO₂ figure, so you can reproduce it live:

$$\begin{aligned}
 t &= L^2 / D \\
 &= (0.4 \text{ m})^2 / (2.0\text{e-}6 \text{ m}^2/\text{s}) \\
 &= 0.16 / 0.000002 \\
 &= 80,000 \text{ seconds} \\
 &= 80,000 / 86,400 \\
 &= 0.93 \text{ days}
 \end{aligned}$$

The ratio heat : CO₂ is **16.7 : 1**. The deck rounds this to “~15×” (see Finding 3). Note that because both times scale as L^2/D , the distance cancels — **the ratio is 16.7 at any separation**, which is a robustness worth mentioning.

IF ASKED TO JUSTIFY THE RATIO ON THE SPOT

“It falls straight out of the diffusivities. Gas in open interstitial air is around 10⁻⁶ m²/s; bulk grain thermal diffusivity is around 10⁻⁷. That is roughly a factor of seventeen, and because both scale with length squared, the ratio is the same whatever separation you pick. The absolute arrival times depend on the geometry you assume — the ratio does not.”

15 Number audit — every quantitative claim traced

This is the section your brief asked for most explicitly. Every number in the deck, where it came from, and what it can bear.

15.1 Numbers that are pure arithmetic from the model

CLAIM	ORIGIN	CLASS
0.9 / 15.4 / 123.5 days (Slide 3)	$t = L^2/D$ with $L = 0.4 \text{ m}$ and our three diffusivities. Reproduced exactly: 0.93, 15.43, 123.46. The arithmetic is unimpeachable; the inputs are estimates.	ENG. EST.
~15× gas advantage (Slide 3)	$15.43 \div 0.93 = 16.7$. Deck rounds down. Harmless but know the real figure.	ENG. EST.
a_w table (Slide 5)	Modified Chung–Pfoest with published ASAE D245 corn coefficients. All five rows reproduce exactly. The equation and coefficients are published; only the choice to use them is ours.	LITERATURE
Onset table 6.7 / 10.8 / 17.4 / 28.9 / 86.8 d	Cardinal-value growth model, our coefficients. All five reproduce exactly.	ENG. EST.
~15 % moisture crossing	Isotherm solved for $a_w = 0.82 \rightarrow 14.66 \text{ % wb at } 30 \text{ }^\circ\text{C}$.	ENG. EST.
1 % DML = 14.67 g CO₂/kg	Stoichiometry: $264 \div 180 \times 10$. Not tunable — it is chemistry.	ESTABLISHED

15.2 Numbers that are simulation outputs

CLAIM	ORIGIN	CLASS
Day 0.8 vs day 42 (Slide 8)	Output of the full simulation on the “deep interior pocket” preset. Verified at hourly resolution: 0.79 and 42.0 . This is a simulation result, not a field measurement — say “in our model” every single time.	ENG. EST.

CLAIM	ORIGIN	CLASS
1.5 / 3.0 / 4.2 σ (Slide 10)	Measured from simulated runs: clean grain peak over 10 days, chosen threshold, hotspot peak. Reproduces exactly.	ENG. EST.
~1,000 ppm depth stratification (Slide 10)	Observed in the simulation between mid-depth and near-face ports. Plausible but not field-verified.	HYPOTHESIS

15.3 Numbers taken from literature

CLAIM	SOURCE AND STATUS	CLASS
2.11–70.94-fold phosphine resistance	Verified exactly. <i>Scientific Reports</i> 13:16497 (2023). LC ₅₀ 0.038–1.277 mg L ⁻¹ . The deck quotes volume and article number correctly — a judge who checks will find it.	LITERATURE
3.89–5.92 % post-harvest loss	NABCONS study for the Ministry of Food Processing Industries, reported to Parliament 2024. Government-commissioned and recent.	LITERATURE
500+ PACS godowns	Press Information Bureau, World's Largest Grain Storage Plan.	LITERATURE
a_w 0.82 growth floor	Supported as a conservative value, NOT as an absolute. Reported minima vary by strain and substrate (see Finding 1). Do not say “cannot”.	LITERATURE
Aflatoxin heat-stable, not removed by drying	Well established in food-safety literature.	ESTABLISHED

15.4 Numbers that are assumptions — own them

CLAIM	WHY WE CHOSE IT, AND WHICH WAY IT CUTS	CLASS
0.4 m spear reach	A practical estimate for a bag spear driven into a stacked wall. Netra owns this number. If it is generous, our comparison looks better; if it is harsh, worse. We believe it is realistic. Requires a citable source or a physical measurement.	ASSUMPTION
14-day inspection interval	Deliberately the <i>generous</i> end — many facilities inspect monthly or slower. Choosing the generous value weakens our own headline, which is the right direction.	ASSUMPTION
60 % chance of finding a pocket in reach	Reflects clustering. Not derived from a study.	ASSUMPTION
Musty-smell detection route	Added deliberately to help the baseline. Without it, conventional inspection never found the pocket at all. This is the honest core of slide 8.	ASSUMPTION
D_{moisture} = 1.5e-8	A single lumped coefficient combining vapour diffusion and convective migration. The weakest of the three transport numbers. Devavrath owns this.	ASSUMPTION
Respiration coefficients R₀, k, Q₁₀	Standard functional form; our coefficients, tuned to reproduce published safe-storage timelines. Krittika owns these.	ASSUMPTION

15.5 What requires validation and cannot be defended today

ITEM	STATUS
That CO₂ actually leads	Our central thesis is modelled, not measured by us. It is supported by storage-technology literature identifying CO ₂ as an earlier indicator, but we have not demonstrated it ourselves.

ITEM	STATUS
temperature in real grain	NEEDS VALIDATION
Any CO ₂ measurement at all	No CO ₂ sensor sourced. Slide 11 states this. NEEDS VALIDATION
Coverage result (N % within X days)	Placeholder on slide 12. Fill it or remove the line. NEEDS VALIDATION
Observed spoilage	Jars sealed ~15 hours. No fungal growth yet, and none expected. NEEDS VALIDATION
Field performance of the probe	No probe lance has been fabricated. All lance geometry is design intent. NEEDS VALIDATION

THE SAFEST WAY TO PHRASE THE CENTRAL CLAIM

Do not say: “CO₂ arrives 15 times faster than heat.” (Sounds measured.)

Say: “Using published transport coefficients, our model puts the gas signal about seventeen times ahead of the thermal one. The physics behind that is not controversial — gas moves through the air gaps, heat has to conduct through insulating grain. What we have not yet done is measure that lead time ourselves in real grain, and that is exactly what our validation programme is for.”

That answer is honest, demonstrates command, and pre-empts the follow-up. It is strictly stronger than the confident version.

Signals

Why CO₂ — asked in every review, and the answer is not “because it is fast”. The complete comparison, including the honest case against.

16 Every candidate signal compared

SIGNAL	INDICATES	SPEED	SENSOR COST	MAIN WEAKNESS
CO ₂	Total respiration — grain, fungal and insect combined	Fast ~1 day	NDIR ₹3–6k	Non-specific. Cannot say <i>what</i> is respiring. Affected by ventilation.
Temperature	Accumulated heat from respiration	Slow ~15 days	Very cheap	Arrives late and attenuated. Confounded by ambient swings.
Relative humidity → a_w	The governing variable for fungal growth	Local, fast	₹250–1,200	Only reports its <i>own</i> location — will not see a pocket a metre away.
Grain moisture	Total water present	Very slow	Meter ₹2.5k	Requires a physical sample. Not continuous. Misses the free/bound distinction.
O ₂ depletion	The mirror of CO ₂ production	Fast	Moderate	Poor signal-to-noise: detecting a small drop from a 21 % baseline is much harder than a rise from 400 ppm.
VOCs / e-nose	Specific fungal metabolites — potentially species identification	Fast	Variable, often high	Drift, calibration burden, cross-sensitivity. Genuinely promising, not yet cheap or robust for a godown.
Visible mould	Established colonisation	Very late	Free	By definition after the fact. Only sees exterior surfaces.
Odour (human)	Advanced deterioration	Very late	Free	Subjective, non-localising, and late.

17 Why CO₂ — and why CO₂ alone is not enough

The case for CO₂

1. **It is upstream.** CO₂ is a direct product of the reaction. Temperature is an *accumulated consequence* of that reaction — heat must build up faster than it dissipates before a probe registers anything.
2. **It travels through the fast medium.** Interstitial air, not insulating kernels.
3. **It integrates all sources.** Grain, fungal and insect respiration all produce it. One signal covers three failure modes.
4. **Its baseline is low.** Rising from ~400 ppm to thousands is a large relative change and therefore easy to detect — unlike O₂, where you hunt a small dip in a large number.

The honest case against — and how we answer it

OBJECTION	OUR ANSWER
CO ₂ is not specific. It cannot tell you whether the source is grain, fungus or insects.	Correct, and we do not claim otherwise. That is precisely why CO ₂ is not used alone. It tells us <i>where</i> and <i>that</i> ; water activity tells us <i>whether fungal growth is possible there</i> ; the combination is what produces a defensible recommendation.

OBJECTION	OUR ANSWER
Ventilation changes CO₂.	Which is why we never use an absolute threshold. Each port is compared against the other lances <i>at its own depth on the same analyser</i> . A ventilation change affects all ports similarly and cancels in the differential.
Ambient CO₂ drifts.	Same answer — differential, not absolute. Plus a rolling per-port baseline.
Sensors drift.	One shared analyser reads all nine ports, so drift is common-mode and cancels in port-to-port comparison. This is the main engineering reason for the shared-analyser architecture , and it is worth stating — most people assume it is only about cost.

THE ANSWER TO “WHY CO₂?” IN ONE PARAGRAPH

“Because it is the earliest signal that physically escapes the pocket. Spoilage is respiration — sugar and oxygen in, CO₂, water and heat out. All three leave together, but CO₂ moves through the air gaps between kernels while heat has to conduct through the kernels themselves, and grain is an insulator. We are not detecting a different event earlier; we are detecting *the same event through the faster channel*.”

Sensors

What is actually inside the probe, how each technology works, and how a sensor survives being buried in dusty, humid grain.

18 CO₂ sensing technologies

TECHNOLOGY	PRINCIPLE	STRENGTHS	WEAKNESSES
NDIR non-dispersive infrared	CO ₂ absorbs infrared at ~4.26 μm. Shine IR through a gas cell, measure how much is absorbed. Beer–Lambert law.	Selective to CO ₂ . Stable over years. No consumable chemistry. Wide range. Mature and affordable.	Costs more than a chemical cell. Needs temperature and pressure compensation. Optics can foul with dust.
Electrochemical	Gas reacts at an electrode; current is proportional to concentration.	Cheap, small, low power.	Consumes reagent — finite life, typically 1–2 years. Drifts. Cross-sensitive. Poorly suited to a multi-year buried deployment.
Metal oxide (MOS)	Gas adsorption changes the resistance of a heated semiconductor film.	Very cheap, robust.	Not specific at all — responds to many gases. Heavy drift, strong humidity dependence. Most cheap “CO ₂ ” MOS modules actually estimate eCO ₂ from VOCs and are not real CO₂ measurements . Unusable here.
Photoacoustic (PAS)	Modulated IR heats the gas; the resulting pressure pulse is heard by a microphone.	Very small, low power, good at low concentrations. Used in modern compact sensors.	Sensitive to external acoustic noise and vibration — a working godown is a noisy place.

RECOMMENDATION, AND THE REASONING

NDIR, single shared analyser, multiplexed across nine ports.

- **NDIR** because it is genuinely selective to CO₂ and stable over years. In buried grain you cannot recalibrate weekly, so drift and consumable chemistry are disqualifying.
- **Wide range matters:** a spoiling pocket in a sealed environment can reach tens of thousands of ppm. A sensor capped at 5,000 ppm saturates and tells you nothing. This is why the deck specifies an SCD30-class part (0–40,000 ppm) rather than a cheaper 5,000 ppm module.
- **Shared, not per-port**, for two reasons — cost, and more importantly **drift cancellation**. Nine separate sensors would each drift independently, and port-to-port comparison would become meaningless. One instrument reading all nine means every comparison is made against the same baseline by the same hardware.

THE FIRMWARE TRAP THAT WOULD SILENTLY DESTROY THE WHOLE PROJECT

Every NDIR sensor ships with **Automatic Baseline Correction (ABC)** enabled. ABC assumes the sensor periodically sees fresh outdoor air at ~400 ppm, and silently re-zeros itself to the lowest value it has seen over a rolling window.

Buried in grain, the sensor never sees fresh air. ABC would therefore progressively re-zero the elevated interstitial CO₂ to “normal” — erasing precisely the signal we exist to detect, while continuing to report confident-looking numbers.

ABC must be disabled in firmware. Slide 9 calls this out, and it is one of the strongest “we have actually thought about this” details in the deck. **Make sure Vishal can explain it** — a judge with embedded experience will recognise it as a real practitioner’s concern rather than a textbook answer.

19 Humidity and temperature sensing

The humidity sensor is arguably more important than the CO₂ sensor, because **relative humidity at equilibrium is water activity** — the variable that actually governs whether fungi can grow.

SENSOR	RH ACCURACY	BUS	VERDICT FOR THIS PROJECT
Sensirion SHT3x	±2 %	I ² C	Chosen. 3.3 V native, well-characterised, and carries an on-chip heater that can burn off condensation — which matters enormously in high-humidity grain headspace.
SHT35 / SHT85	±1.5 %	I ² C	Better, more expensive. Worth it if a _w precision becomes limiting.
BME280	±3 %	I ² C	Too coarse. ±3 % RH is ±0.03 a _w , and we are distinguishing 0.82 from 0.85.
DHT22	±2–5 %	1-wire	Rejected. Drifts, slow, and known to fail in condensing conditions — exactly our environment.

WHY ACCURACY MATTERS MORE HERE THAN ANYWHERE ELSE IN THE SYSTEM

RH accuracy converts directly into a_w accuracy. ±2 % RH is ±0.02 a_w. Our two decision thresholds are 0.82 and 0.85 — only 0.03 apart. So sensor error is a meaningful fraction of the decision margin. **This is an honest limitation worth volunteering**, and the answer is that we treat 0.82 as a conservative trigger rather than a precise boundary, and we require the CO₂ anomaly to agree before escalating.

20 Protecting a sensor inside grain

The sensor must be **in gas communication with the interstitial air** while being protected from dust, liquid water, condensation and mechanical damage. These requirements conflict, and resolving them is real engineering.

THREAT	CONSEQUENCE	MITIGATION
Grain dust	Fouls NDIR optics; blocks the sampling path; buries the humidity element	Sintered bronze or stainless filter at the port; hydrophobic PTFE membrane over the sensor cavity; larger settling volume before the optical cell
Condensation	Liquid water on a humidity sensor pins the reading at ~100 % RH and is the single most likely field failure	Hydrophobic membrane; the SHT3x on-chip heater pulsed periodically to drive off condensate; reject readings taken during a transient
Liquid ingress	Destroys electronics	Electronics potted and isolated from the gas path; the gas path itself is drained and sloped; IP-rated head unit above the stack

THREAT	CONSEQUENCE	MITIGATION
Mechanical load	Bag weight and insertion force bend or crush the lance	Structural tube sized for buckling under insertion load; sensor cavity recessed so it never bears load
Dead volume	Stale gas in the tubing gives you last hour's reading — a subtle and serious error	Minimise tube volume; purge each port before dwelling on it; wait for a stable reading before recording

THE DESIGN PRINCIPLE

Separate the gas path from the electronics path. Gas is sampled at depth and brought to an analyser that lives in a protected head unit; only rugged, potted digital sensors go down the lance. This is why the shared-analyser architecture is not merely a cost decision — **it also removes the most fragile component from the hostile environment entirely.**

Electronics

21 ESP32 — what it is and why it fits

The ESP32 is a low-cost microcontroller with integrated Wi-Fi and Bluetooth. A **microcontroller** is a complete small computer on one chip — processor, memory and input/output together — designed to run one program continuously and talk to hardware directly.

The peripherals that matter here

FEATURE	WHAT IT IS	OUR USE
GPIO	General-purpose pins, individually readable or drivable	Valve and pump control, status LEDs
I²C	Two-wire bus; many devices share one pair of wires, each with an address	The main bus. SHT3x sensors, the CO ₂ analyser, the OLED, the multiplexer
UART	Simple serial link	Alternative CO ₂ sensor interface; debug console
SPI	Faster synchronous bus	SD card for offline logging
ADC	Converts an analogue voltage to a number	Battery monitoring. Note: the ESP32 ADC is known to be non-linear — we avoid depending on it for measurement
Wi-Fi / BLE	Integrated radios	Opportunistic upload; BLE for local field configuration
Deep sleep	Sub-milliamp standby with timed wake	Critical for battery life — we sample periodically, not continuously

Why ESP32 rather than the alternatives

OPTION	WHY NOT
Arduino Uno (ATmega328P)	No networking, 2 KB RAM, 8-bit. Cannot buffer a dataset or run the interpretation locally.
Raspberry Pi	Runs a full OS — needs clean shutdown, corrupts SD cards on power loss, draws far more current. Overkill and less reliable for a sensing node that must survive unattended.
STM32	Excellent silicon, but no integrated radio and a steeper toolchain. Slower to build with, for no advantage at this stage.
ESP32	Chosen. Integrated Wi-Fi/BLE, ample RAM, dual I ² C controllers, deep sleep, 3.3 V logic matching all our sensors, ~₹500, and an enormous library ecosystem. It does the job with the fewest additional parts.

THREE HARDWARE FACTS YOU MUST BE ABLE TO STATE

- 1. ESP32 GPIO is not 5 V tolerant.** Everything in our design is 3.3 V I²C deliberately, so no level shifters are needed and nothing gets destroyed by a wiring slip.
- 2. The SHT3x has only two I²C addresses (0x44 / 0x45).** Three sensors on one bus collide. Solved either with a **TCA9548A multiplexer** (as slide 12 says) or by using the ESP32’s **two independent I²C controllers**. Knowing both routes is better than knowing one.
- 3. Wi-Fi and the analyser peak together.** Power supply must be sized for the simultaneous peak, not the average, or the node browns out during transmission — an intermittent fault that is miserable to diagnose.

22 End-to-end system architecture

Sensors

3× SHT3x at depth (I²C) · one shared NDIR analyser fed by a manifold · optional contact acoustic sensor



I²C at 100 kHz, 2.2 k pull-ups for long cable runs

ESP32 node — one per stack

Sequences the manifold, purges and dwells on each port, reads all sensors, **CRC-checks every frame**, timestamps, and computes the local anomaly score. Buffers to flash when offline.



MQTT over Wi-Fi when available; store-and-forward when not

Gateway / backend

Ingests readings, writes to a time-series store, runs the interpretation layer: isotherm, growth kinetics, risk scoring, ranking.



REST API

Dashboard and alerts

Stack-level risk map, per-port history, and the action card. Alerts by SMS or app because a godown manager is not watching a screen.

OFFLINE-FIRST IS A DESIGN REQUIREMENT, NOT A FALLBACK

Slide 9 says “runs entirely with Wi-Fi off”. Rural godown connectivity cannot be assumed, so **the decision logic must never depend on a network**. The node computes its own anomaly score and can raise a local alarm unaided; synchronisation is opportunistic. This is also why the demo software has zero external dependencies — the same principle, applied to the presentation.

Mechanical

Devavrath's section. The probe is the part of this project that is genuinely novel, and it is also the part most likely to be dismissed as “a sensor in a pipe” if described carelessly.

23 Realistic probe-lance design

PARAMETER	DESIGN INTENT	REASONING
Length	2.0–2.5 m usable	Must reach the thermal and diffusive centre of a stack, which is the whole point.
Outer diameter	20–25 mm	Trade-off: thin enough to slip between bag courses without cutting sacks, thick enough to resist buckling under insertion load.
Material	Stainless 304, or aluminium for the prototype	Corrosion resistance in humid grain; food-contact acceptability; stiffness. Aluminium is cheaper and lighter for a first article.
Tip	Conical, closed, hardened	Parts grain rather than coring it, and prevents the lance filling with grain during insertion.
Ports	Rings of small perforations at 0.5 / 1.2 / 2.0 m	Several small holes rather than one large one — samples a representative volume and resists blocking by a single kernel.
Filtration	Sintered element + hydrophobic PTFE membrane	Passes gas, stops dust and liquid water. The membrane is what keeps condensation off the sensing element.
Internal routing	Separate gas lumen and cable lumen	Keeps the sampled gas away from the wiring. Also allows the gas lumen to be purged without disturbing sensors.
Port isolation	Internal seals between depth zones	Essential. Without them the lance short-circuits the very depth gradient we are trying to measure — every port would read the same mixed gas.
Head unit	Sealed enclosure above the stack	Houses the analyser, pump, manifold and ESP32 in a protected, serviceable location.

THE DETAIL THAT PROVES YOU HAVE ACTUALLY THOUGHT ABOUT IT

Inter-port sealing. If the three depth ports share one open internal volume, gas from all three mixes inside the tube and every port reads the average. The depth resolution — the entire basis of the depth-matched differential detector — would be destroyed by a manufacturing shortcut that looks harmless.

If Devavrath raises this unprompted, it demonstrates real design thinking. It is exactly the kind of failure that separates a considered prototype from a pipe with holes in it.

24 The mechanical brief, and where CFD genuinely helps

What Devavrath owns

- **CAD** of the lance, tip, port geometry, seals and head unit
- **Material selection** with justification — corrosion, stiffness, food contact, cost
- **Insertion mechanics** — estimating required force, and whether one person can do it

- **Structural analysis** — buckling under axial insertion load; bending under bag pressure
- **Sensor-chamber design** — volume, purge time, dead volume
- **Manufacturability** — can this be made in a college workshop, and later at scale

Where CFD is genuinely useful — and where it is decoration

PROPOSED SIMULATION	VERDICT
Sensor-chamber purge / dead-volume flow	Genuinely useful. Answers a real design question with a real number: how long must the pump dwell on a port before the reading reflects that port rather than the previous one? This directly sets the manifold sequencing timing in firmware.
CO2 diffusion from a pocket through porous grain to a port	Useful. This is the model that produced the headline timing. Running it properly in CFD with a porous-medium formulation would upgrade the claim from ENG. EST. toward a defensible simulation result.
Effect of port geometry on sampling volume	Moderately useful. Helps choose perforation pattern. Lower priority.
Full-stack 3D transient CFD of an entire godown	Not useful. Enormous computational cost, dominated by boundary conditions and porosity assumptions you cannot verify. It would produce a colourful picture and no additional confidence. Do not do this for presentation points.

WHAT CFD CAN AND CANNOT PROVE — SAY THIS IF ASKED

CFD can show: that under stated assumptions about porosity, tortuosity and boundary conditions, a gas signal reaches the port in roughly this time; how long a chamber takes to purge; how geometry changes the sampled volume.

CFD cannot prove: that real grain has the porosity you assumed; that a real pocket respire at the rate you assumed; that the real thing works. **A simulation validates a model, not reality.** Only the jar and column experiments can close that gap.

Saying this yourself is far stronger than being told it. It signals you know what simulation is for.

Biotechnology

Krittika and Netra’s section. Their sign-off is what makes every biological number in the deck defensible rather than merely typed.

25 Fungal biology for this project

Why *Aspergillus flavus* specifically

- **It is the canonical aflatoxin producer** on maize — the classic pairing in the literature, and a documented Indian problem.
- **It is a storage fungus, not a field fungus** — it grows at the relatively low water activities found in stored grain, which is exactly our regime.
- **Its cardinal values are published**, so a growth model can be built on citable numbers rather than invented ones.

The cardinal-value model, in plain terms

A cardinal model says an organism has a minimum, an optimum and a maximum for each environmental variable, and growth rate falls off from the optimum towards the limits. Our model uses:

VARIABLE	MIN	OPTIMUM	MAX
Temperature	12 °C	33 °C	43 °C
Water activity	0.82	0.99	—

Growth rate is the product of a temperature factor and a water-activity factor, scaled by a maximum rate. Days-to-onset is then simply the reciprocal. **ENGINEERING EST.** — standard approach, our coefficients.

KRITTIKA’S CORE EXPLANATION, IN ONE PARAGRAPH

“Fungi do not respond to how much water is present. They respond to how much is *available*. Some water in a kernel is chemically bound to starch and protein and no organism can access it. Water activity is the free fraction, on a scale of zero to one. Below about 0.82, *A. flavus* cannot extract enough water to grow — and because relative humidity at equilibrium *is* water activity, we can measure the governing variable with an ordinary humidity sensor.”

26 Safe validation experiments

SAFETY BOUNDARY — NON-NEGOTIABLE

Do not deliberately culture or inoculate aflatoxigenic *A. flavus* outside appropriate laboratory containment, and never for a demonstration. Aflatoxin is a potent carcinogen and the spores are a respiratory hazard.

Everything below uses either **naturally occurring colonisation** of wetted grain (no inoculation) or **abiotic proxies**. All wet-grain work belongs in the lab, under the mentor's supervision, with gloves and a dust mask, and jars stay sealed outside the lab.

Experiment 1 — water activity, the one already running

Question	Does our sensing chain measure water activity correctly, and does the decision engine respond correctly?
Design	Two sealed jars of maize, identical but for moisture. Control at storage moisture; test wetted to ~18 % wb.
Measurement	SHT3x in the headspace of each; reading typed into the same decision-engine code the simulation uses.
Expected	Control below 0.82; test above it, with a predicted onset. HYPOTHESIS
Control for	Same grain, mass, container and temperature. Moisture is the only variable.
Limitation	Demonstrates a_w measurement only — not spoilage, not CO ₂ .

Experiment 2 — sensor calibration against salt standards

Saturated salt solutions hold a known water activity in the air above them. This is the standard method and costs almost nothing:

SALT (SATURATED)	A_w AT 25 °C	ROLE
Sodium chloride	0.753	Below our threshold
Potassium chloride	0.843	Straddles the 0.82 floor
Potassium sulphate	0.973	Above

Why this is worth doing: it converts “our sensor says 0.87” into “our sensor is calibrated against standards and says 0.87.” A judge asking “how do you know your sensor is accurate?” is asking a fair question, and “it is a good sensor” is not an answer.

Experiment 3 — the CO₂ lead-time test (the one that matters most)

Question	Does a developing moisture event produce a detectable CO₂ rise before a detectable temperature rise? This is the project's central unvalidated claim.
Design	A grain column (a bucket or drum) with a wetted pocket introduced at a known depth. CO ₂ , RH and temperature logged at two or three depths continuously.
Control	An identical column with no wetted pocket, logged simultaneously.
Interval	Every 15–60 minutes, continuously, for 2–4 weeks.
Acceptance	CO ₂ in the test column departs from the control by a defined margin measurably earlier than temperature does.
Limitations	A column is not a bag stack. Boundary conditions differ. Single commodity, single event.
Status	NEEDS VALIDATION — not yet run. Requires the CO ₂ sensor.

EXPECTED RESULT VERSUS DEMONSTRATED RESULT

Never blur these. Today you have a demonstrated water-activity measurement and an expected CO₂ lead time. Saying “we expect” where you mean expect costs you nothing and protects everything else you claim. Saying “we found” where you mean expect is the single fastest way to lose a technical review.

Software and intelligence

Slide 10 makes a claim most teams would not dare make: that we deliberately chose *not* to use a black box. This part explains why that is the stronger engineering position.

27 Software architecture and the CS division of work

LAYER	RESPONSIBILITIES	OWNER
Firmware	Manifold sequencing, purge-and-dwell timing, sensor acquisition, CRC validation , timestamping, local anomaly computation, deep-sleep power management, offline buffering, OTA update	Vishal
Backend	MQTT ingest, time-series storage, the interpretation pipeline (isotherm → kinetics → risk), REST API, alert dispatch, tamper-evident hash-chained log	Mohit
Frontend & analytics	Stack risk map, per-port time series, action cards, the detector's explainability view, and the digital-twin demo itself	Chiranjib

ONE IMPROVEMENT ON THE SPLIT IN YOUR BRIEF

Your draft put anomaly detection with the dashboard. **Move the detector into the firmware layer and give it to Vishal**, alongside the noise model. The reason: the detector must run **offline on the node**, because slide 9 promises the system works with Wi-Fi off. A detector that lives only in the dashboard breaks that promise. The dashboard should *visualise* the detector, not *be* it.

Data flow, concretely

Every hour, per node:

```

for port in 1..9:
    switch manifold to port
    run pump PURGE_MS          # clear dead volume - critical
    wait for reading to stabilise
    read CO2, RH, temperature
    verify CRC                  # reject corrupt frames, never guess
    store {port, t, co2, rh, temp}

compute depth-matched anomaly for all 9 ports
if any port >= 3.0 sigma for 3 consecutive cycles:
    raise local alert          # works with no network
publish batch via MQTT if link available, else buffer to flash

```

28 The detector, and why it is not a threshold

Slide 10 rejects two obvious approaches before arriving at the third. Being able to explain *why* each was rejected is worth more than the final answer.

	APPROACH	WHY REJECTED OR CHOSEN
×	Fixed absolute threshold “alert above 3,000 ppm”	Godown temperature swings seasonally, and so does baseline respiration. Any absolute limit either cries wolf in summer or sleeps through winter . There is no single correct value.
×	Compare all nine ports together	Depth stratification is real — roughly 1,000 ppm between mid-depth and near-face ports, because deep grain is furthest from a venting face. Pooling all nine buries a genuine local anomaly inside that natural spread.
✓	Depth-matched differential	Each port is compared only against the other lances at its own depth , on the same analyser. Cancels sensor drift and slow ventilation changes simultaneously, because both affect all ports at that depth equally. This is what lets a real pocket stand out.

For each port i at depth d :

```

peers      = other ports at the SAME depth d
mu, sigma  = mean and spread of peers
z_i        = (reading_i - mu) / sigma

```

ALERT if $z_i \geq 3.0$ for 3 consecutive readings

Measured separation in simulation:

```

clean grain peaks at    1.5 sigma
alert threshold         3.0 sigma
real hotspot reaches    4.2 sigma

```

WHY “THREE CONSECUTIVE READINGS” MATTERS

A single noisy sample must never raise an alarm. Requiring persistence trades a small amount of latency (three hours at hourly sampling) for a large reduction in false alarms. **In a system whose value depends on being trusted, a false alarm is more expensive than three hours.**

Threshold vs anomaly vs prediction — we use all three, at different stages

METHOD	WHERE WE USE IT	WHY THERE
Threshold	The $a_w = 0.82$ growth floor	Appropriate because the threshold is biological, not statistical . It comes from the organism, not from our data.
Anomaly detection	The CO ₂ differential	Appropriate because there is no correct absolute value — only “different from its neighbours”.
Predictive model	Days-to-mould-onset	Appropriate because we are forecasting the crossing of a known threshold, which is a genuine prediction problem.

DO NOT CALL THIS AI

There is no machine learning in this system, and that is a **deliberate and defensible choice**, not a shortfall. Slide 10 says so explicitly: “*an explainable detector we can defend beats a black box we cannot.*”

If a judge asks “where is the AI?”: “There isn’t any, on purpose. We have nine sensors and no labelled failure dataset — training a model on that would produce something we could not explain and could not defend. Every alert we raise carries its own reasoning: this port, this many sigma above its depth peers, this water activity, this many days. When we have a labelled dataset from the validation programme, machine learning becomes worth revisiting. Today it would be decoration.”

That answer scores **better** than claiming AI, because it demonstrates you know what ML requires and when it is inappropriate.

29 Risk scoring and sensor fusion

No single signal is sufficient. CO₂ says something is respiring but not what. Water activity says growth is possible here but not that it is happening. Temperature confirms late. **Combining them produces confidence none of them has alone.**

```
risk = 100 x ( 0.35 x aw_score
              + 0.40 x urgency_score
              + 0.25 x co2_score )

aw_score      = normalised distance above the 0.82 floor
urgency_score = 1 - (days_to_onset / 30), clamped to 0..1
co2_score     = normalised sigma anomaly, clamped to 0..1

ALERT when risk >= 45 OR co2 sigma >= 3.0 sustained
```

BAND	RISK	MEANING AND ACTION
GREEN	0–24	Normal. a_w below the floor, no anomaly. Log only.
YELLOW	25–44	Watch. One signal elevated. Increase sampling rate on that port.
ORANGE	45–69	Investigate. Two signals agree. Issue an action card; Tier 1 aeration.
RED	70–100	Urgent. a_w above the floor with a short predicted onset and a confirmed anomaly. Escalate.

WHY WEIGHTED-SUM RATHER THAN SOMETHING CLEVERER

Because **every term is inspectable**. When an alert fires you can say exactly which component drove it and by how much. A neural network on nine sensors would be both unjustifiable (no training data) and unexplainable (no reasoning to show the operator). The weights themselves are an **ASSUMPTION** — chosen, not derived — and are on Vishal’s verification sheet.

Competition

30 Existing technology, rigorously compared

THE RULE FOR THIS SECTION

Never claim existing systems cannot do something they can. Commercial grain monitoring is a mature field with capable products. Overstating their limitations is the fastest way to look naive to a judge who knows the market. Our argument is about **geometry, price point and output** — not about competitors being incapable.

APPROACH	WHAT IT DETECTS	WHERE IT WORKS	WHERE IT FAILS <i>FOR US</i>
Manual inspection	Visible mould, staining, insect holes	Exterior surfaces; skilled and cheap	Cannot see the interior at all
Spear sampling	Grain condition in the sampled volume	Within ~0.4 m of a face	Cannot reach the centre; non-uniform problems; misses concealed life stages
Moisture meters	Total moisture of a physical sample	Intake QC, spot checks — accurate and standard	Requires a sample, so cannot be continuous or interior; measures total, not available, water
Temperature cables	Interior temperature at fixed points	Genuinely effective in silos , standard practice, well proven	Designed for permanent vessels; heat arrives late through insulating grain; awkward and uneconomic in stacks that get rebuilt
Thermal imaging	Surface temperature distribution	Fast, non-contact exterior survey	Sees surfaces only. A 1.5 m deep pocket produces essentially no surface signature until very late
Handheld CO₂ meters	CO ₂ at a point, at one moment	Spot checks; the underlying signal is the right one	Not continuous, no depth resolution, no baseline history — so no anomaly detection is possible
Electronic nose	VOC signatures, potentially species-specific	Research settings; genuinely promising	Drift and calibration burden; cost; not yet field-robust for godowns
Commercial IoT (TeleSense, OPI etc.)	Temperature, humidity, sometimes CO ₂ , with cloud analytics	Technically capable and well engineered. Standard in large Western operations	Designed and priced for bulk silos , not bagged stacks; economically out of reach for a PACS-level facility

31 Novelty versus engineering integration

Judges reward teams who can separate these honestly. Claiming novelty for something standard is the fastest way to lose credibility.

COMPONENT	STATUS	NOTE
CO ₂ as a spoilage indicator	Established	Documented in storage literature. We do not claim to have discovered this.
Sorption isotherms for a_w	Established	Published equations, published coefficients, used as intended
Predictive mycology models	Established	Standard cardinal-value approach
NDIR sensing	Established	Commodity technology
Interior-resolved sensing designed for bag stacks	NOVEL	Commercial monitoring targets silos. Bag stacks have different access constraints and different diffusion behaviour. The lance is designed for that geometry.

COMPONENT	STATUS	NOTE
Single-analyser multiplexed depth mapping	NOVEL	Makes depth-resolved gas mapping affordable and eliminates inter-sensor drift, so every depth comparison is made by one instrument against one baseline.
Prediction rather than measurement	NOVEL	Output is a modelled time-to-risk with a deadline, not a sensor reading. Converts monitoring into prevention.
Closed-loop verification	NOVEL	Post-treatment signature decay confirms efficacy — addressing a failure mode current practice does not detect at all.

THE ONE-SENTENCE NOVELTY CLAIM

“The individual signals are established science. What is new is the combination applied to bag-stack geometry, the single-analyser architecture that makes depth-resolved gas mapping affordable, and the shift from reporting a measurement to issuing a dated, evidence-backed prediction.”

Evidence, cost and failure

32 What is proven versus what is not — today

CLAIM	STATUS	BASIS
Two sealed maize jars, one at storage moisture, one wetted	YES	Physically prepared and sealed
SHT3x reading headspace on the same ESP32 firmware built for the field probe	YES	Hardware built, firmware flashed, CRC-checked
Reading fed into the same decision-engine code as the simulation	YES	One codebase, one source of truth
Dry jar below 0.82; wet jar above, with a predicted onset	YES	Measured water activity
Working digital twin with real finite-difference physics	YES	Self-checking, headless assertions pass before every session
“We watched grain spoil”	NO	Sealed ~15 hours; fungus needs days
“We measured CO ₂ ”	NO	Sensor not sourced
“CO ₂ leads temperature in real grain”	NO	Modelled and literature-supported; not measured by us
Coverage result (N % within X days)	NO	Analysis not yet run

33 Cost and scalability

Prototype bill of materials

COMPONENT	QTY	PURPOSE	DECK PRICE	ACTUAL PAID
SCD30 NDIR CO ₂ analyser	1	The core gas measurement, 0–40,000 ppm	₹5,000	not yet
SHT3x humidity/temperature	3	Water activity at three depths	₹3,000	₹750
ESP32 DevKit	2	Node controller plus a spare	₹1,200	₹600
TCA9548A multiplexer	1	Resolves the SHT3x address collision	₹250	₹250
Supply, cable, connectors	—	Power and wiring	₹1,000	₹1,044
Total			₹10,450	₹8,200

Both figures sit inside the ₹19,800 proposal budget. See Finding 4 — the deck’s SHT3x line is four times what you actually paid.

Scaling economics

SCALE	CONFIGURATION	INDICATIVE COST	NOTE
1 probe assembly	3 lances, 9 ports, 1 analyser, 1 node	~₹10,000	Prototype, hand-built

SCALE	CONFIGURATION	INDICATIVE COST	NOTE
10 units	10 stacks in one godown	~₹80,000	Component discounts begin; lance fabrication amortises
100 units	A storage circle	~₹6–7 lakh	Volume pricing; the analyser dominates unit cost

ENGINEERING EST. — these are scaled estimates from component pricing, **not quotations**. Say “on the order of” rather than quoting them as firm.

THE DEPLOYMENT PATH

1. **Laboratory** — jars and columns, controlled conditions
2. **Instrumented test stack** — a real bag stack with a known induced pocket
3. **Single godown pilot** — documented lead-time and chemical-use comparison against that facility’s existing practice
4. **Storage circle** — extension across a district
5. **PACS-level deployment** — the 500+ new godowns with no instrumentation

34 Failure modes (FMEA)

FAILURE	CAUSE	EFFECT	DETECTION	MITIGATION
ABC re-zeroing	NDIR auto-calibration left enabled	Signal silently erased — worst case in the system	Baseline drifts down over days with no physical cause	Disable ABC in firmware ; periodic reference check
Sensor drift	Ageing optics or electronics	Slow bias in readings	Port-to-port disagreement grows	Shared analyser makes drift common-mode; differential comparison cancels it
Condensation	High humidity, temperature swing	RH pinned near 100 %; false high a_w	Reading saturates and stops responding	Hydrophobic membrane; SHT3x on-chip heater; reject transients
Dust blockage	Grain fines migrating to the port	Port stops responding; looks falsely normal	Reading goes flat while peers vary	Sintered filter; periodic purge; flat-line detection in firmware
False positive	Noise spike, local disturbance	Unnecessary intervention; trust erodes	Anomaly does not persist	3-consecutive-reading rule ; require two signals to agree before escalating
False negative	Pocket between lances, beyond diffusive reach	Missed event	Only found later by inspection	Lance spacing informed by the coverage analysis; honest statement of coverage limits
Ventilation change	Doors opened, aeration run	All CO ₂ readings shift together	Whole-facility shift, not one port	Differential comparison cancels common-mode change
Network loss	No rural connectivity	No upload	Gateway sees no data	Offline-first by design ; node buffers and alerts locally
Power failure	Battery flat, supply interrupted	Node dead	Missed heartbeat	Deep sleep duty-cycling; heartbeat monitoring; solar option
Probe damage	Insertion force, handling	Leaks, blocked lumen	Port reads implausibly like ambient	Structural sizing; ambient-match detection; field-replaceable lance
Grain variety change	Different commodity, different isotherm	a_w conversion wrong	Not self-evident — dangerous	Coefficients parameterised per commodity; refuse to report a_w for an unconfigured commodity

Team and delivery

35 Responsibility matrix

MEMBER	OWNS	MUST BE ABLE TO ANSWER
Chiranjib CSE, lead	System architecture, integration, decision engine, dashboard, the digital twin	How the whole loop fits together; what is built versus proposed; why an explainable detector rather than ML; the cost case
Mohit CSE	Transport physics and numerical modelling; backend and data pipeline	Why gas outruns heat; Fick and Fourier; the stability condition of the solver; where the 0.9 / 15.4 / 123.5 figures come from and what class of claim they are
Vishal CSE	Firmware, sensing pipeline, noise model, the detector	Why noise is simulated at all; why depth-matched differential rather than a threshold; why ABC must be disabled ; the 1.5 / 3.0 / 4.2 sigma separation
Krittika Biotech	All biological numbers. Water activity, growth thresholds, respiration coefficients, the isotherm choice	Why a_w and not moisture; where 0.82 comes from and that it is a conservative end of a range ; why aflatoxin is irreversible; why maize
Netra Biotech	The inspection baseline and its fairness; entomology; validation protocol design	Why 14 days; why 0.4 m; why we added the musty-smell route ; why the comparison is not rigged — the most likely attack surface in the whole review
Devavrath Mechanical	Probe geometry, materials, insertion mechanics, transport constants, CFD	Lance dimensions and why; inter-port sealing ; whether $D_{moisture}$ is molecular or effective (it is effective); what CFD can and cannot prove

THE RULE THAT PROTECTS THE WHOLE TEAM

When a judge asks where a number came from, the person who owns it answers — not the person who typed it. Every biological threshold is defended by Krittika, every baseline assumption by Netra, every transport constant by Devavrath. That division is not cosmetic: it is the difference between a team that can defend its model and one that can only run it.

But **every member must be able to give the 30-second version of the whole system.** A judge who asks the mechanical student “so what does your project do?” and gets only probe geometry has learned something bad about the team.

36 Slide-by-slide speaking guide

All fourteen slides. For each: what it says, what it means, what to say, where to pause, what not to claim, and where the judge is most likely to interrupt.

Slide 1 — ANNA DRISHTI / See Deeper. Save Grain.

What it says	Title, tagline “DETECT · PREDICT · PROTECT”, and the positioning line “Deep-Stack Grain Monitoring & Early-Warning System”.
What it means	The three verbs are the actual architecture, in order. Detect is the sensing; Predict is the growth model; Protect is the tiered intervention.
10 seconds	“Anna-Drishti — a system that sees inside a grain stack and warns before the damage is done.”
30 seconds	“Indian grain sits in sacks stacked metres deep. Damage starts in the middle, where nobody can look. Anna-Drishti puts probes inside the stack and detects spoilage while it is still reversible — detect, predict, protect.”
Do not	Do not linger. This slide earns nothing. Ten seconds maximum.

Slide 2 — The Problem / Spoilage starts where inspection stops

What it says	Three-step progression — localised moisture, hidden deterioration, silent spread — plus the NABCONS 3.89–5.92 % figure.
What it means	The problem is not that grain spoils. It is that it spoils where nobody can observe it , so the first evidence arrives after the damage.
60 seconds	“Three things happen in sequence. A roof leak or a wet sack creates a small damp pocket inside a stack. That raised moisture makes conditions favourable for fungal growth — and in the case of <i>Aspergillus flavus</i> , aflatoxin risk can develop before any mould is visible. And because conventional inspection can only sample what it can reach, the damage spreads silently. India’s post-harvest cereal losses were assessed at 3.89 to 5.92 percent in a NABCONS study reported to Parliament in 2024.”
Point at	The three numbered blocks, in order. Then the NABCONS figure.
Pause	After “before any mould is visible.”
Judge may ask	“Where is that loss figure from?” — answer: NABCONS study for the Ministry of Food Processing Industries, reported to Parliament in 2024. Government-commissioned, not an industry estimate.
Do not	Do not claim the whole 3.89–5.92 % is storage loss. It is total post-harvest loss across all stages. Saying so yourself is a credibility gain.

TYPO ON THIS SLIDE

“DETORIATION” appears twice — should be “**DETERIORATION**”. Fix before presenting. A spelling error on the problem slide is a small thing that costs disproportionately.

Slide 3 — The Insight / Spoilage breathes, gas arrives first

What it says	The three arrival times over 0.4 m — CO ₂ 0.9 days, heat 15.4 days, moisture 123.5 days — and “gas leads by ~15×”.
What it means	This is the intellectual core of the entire project. One biological event; three products; three transport mechanisms; three arrival times. We stand where it arrives first.
10 seconds	“Spoiling grain breathes. The gas gets out long before the heat does.”
60 seconds	“Grain is alive, and so are the fungi on it. Both consume sugar and oxygen and release CO ₂ , water and heat — from a single reaction. But those three products travel very differently. CO ₂ escapes through the air gaps between kernels. Heat has to conduct through the kernels themselves, and grain is an excellent insulator. Over the same forty centimetres, our model puts the gas at under a day and the heat at over fifteen. The same biological event announces itself as gas about two weeks before it announces itself as warmth.”
Point at	The three arrival times, in order. Let the contrast do the work.
Pause	After “two weeks before it announces itself as warmth.” This is the single most important sentence in your presentation.
Judge will ask	“ Where do those numbers come from? ” This is the highest-probability interruption in the deck. See §15 and Killer Question 3.
Do not	Do not say “we measured”. Say “our model gives”. The numbers are computed from published diffusivities, not observed by you.

Slide 4 — The Solution / Sense → Interpret → Decide → Verify

What it says	The four-stage loop, and the framing line “from ‘something smells’ to ‘L2 at 2 m, act in 6 days’”.
What it means	The output is a decision , not a reading. And the loop closes — verification is the stage nobody else has.

60 seconds	“Four stages. We sense interstitial CO ₂ , humidity and temperature at nine ports across three depths. We interpret — humidity becomes water activity, and water activity plus temperature becomes a growth forecast. We decide — a composite risk score, ranked ports, and an action card with a deadline. And then we verify: after the intervention, the signature should decay. If it does not, we escalate rather than assume success. Current practice leaves this loop open at both ends.”
Emphasise	“The output is not an alarm. It is a port, a reason, and a deadline.”
Judge may ask	“Does anything else verify?” — No. A failed fumigation and a successful one look identical today; the difference surfaces weeks later when infestation returns.

Slide 5 — The Biology / Moisture is the total, a_w is the spendable part

What it says	The moisture–temperature– a_w table, the 0.82 and 0.85 thresholds, the 25–35 °C toxin band, the time-to-onset table, and why maize.
What it means	There is a specific, measurable threshold , which is what makes prediction possible at all. Without a threshold there is nothing to forecast the crossing of.
Krittika should deliver this slide.	A second voice resets attention, and this is her domain. It also demonstrates the team structure rather than asserting it.
60 seconds	“Not all water in a kernel is usable. Some is chemically bound to starch and protein and fungi cannot touch it. Water activity is the free fraction, on a scale of zero to one. Below about 0.82, <i>A. flavus</i> growth is not supported. Above it, the clock starts — and the clock is what we report. Notice the table has both moisture and temperature columns: two batches at the same moisture behave differently at different temperatures, because temperature changes how much of that water is free.”
Point at	The 0.82 line first, then sweep down the onset table.
Do not	Do not say “physically cannot”. See Finding 1. Say “is not supported in the literature we rely on, and 0.82 is the conservative end of a published range.”
Judge may ask	“Why maize?” — major stored cereal in the godown system; published ASAE D245 coefficients exist for shelled corn so the conversion is a standard, not an invention; and it is what is in our jars. One commodity validated properly rather than four claimed loosely.

Slide 6 — The Biology / You can stop the fungus, you cannot unmake the toxin

What it says	Aflatoxin is a liver carcinogen, heat stable, not destroyed by drying or cooking. “We do not measure toxin.”
What it means	This slide exists to state a limitation before a judge finds it . It is the highest-credibility slide in the deck.
60 seconds	“Aflatoxin is a liver carcinogen. Drying does not destroy it. Cooking does not destroy it — it is heat stable. Once it is in a consignment, that grain’s fate is decided. So a system that tells you grain is already contaminated has told you nothing you can act on. The only intervention that changes the outcome happens before the fungus grows — which makes this an inherently predictive problem. And to be explicit: we do not measure toxin. We predict the conditions that create it, and we say so in every output the system produces.”
Pause	After “that grain’s fate is decided.”
Do not	Never skip this slide to save time. Volunteering the limitation is worth more than anything you would say instead.

Slide 7 — Current Practice / The problem isn’t diligence, it’s reach

What it says	How inspection works, its three physical limits, the spear-reach quote, and the phosphine resistance data.
What it means	The framing is everything: this is not a criticism of inspectors . It is geometry.
60 seconds	“An inspector walks the stack looking for staining and insect holes, then pushes a hollow spear into some bags. Careful, skilled work — limited in three ways. It only reaches the outside; a spear reaches about forty centimetres and a stack is metres deep. Problems are not spread evenly — a clean sample here says nothing about a pocket two metres away. And the early stages are invisible, because eggs and larvae develop inside the kernels. A spear reaches forty centimetres. Our hotspot is at one and a half metres. No amount of diligence gets a spear to that location.”
Pause	After “to that location.” Then: “This is not a criticism of inspection. It is precisely the reason an interior probe needs to exist.”
Judge may ask	“Isn’t 1.6 m arbitrary?” — See Finding 2. Know the four face distances.
Strong detail	The resistance figures are exact and verifiable — 2.11 to 70.94-fold, <i>Sci. Reports</i> 13:16497, 2023. Quoting a volume and article number correctly signals real scholarship.

Slide 8 — The Evidence / We weakened our own headline number on purpose

What it says	The five baseline assumptions, why each is generous to conventional inspection, and the day 0.8 vs day 42 result.
What it means	This is the most sophisticated slide in the deck. Most teams inflate their comparison. You handicapped yours and said so.
Netra should deliver this slide.	She owns the baseline and will field the fairness question anyway.
60 seconds	“Here is how we modelled conventional inspection, and why every assumption is generous to it. Fortnightly inspection — many facilities are monthly or slower. Spear reach 0.4 m, the physical limit. Sixty percent chance of finding a pocket that is in reach. We assume inspectors never miss visible mould, not once. And we added a second detection route — musty smell and warm bags — specifically to help the baseline. Our first model had conventional inspection never finding this pocket at all, which is arguably what the guidance implies. We thought that looked rigged. That is where day 42 comes from. Without it, our gap would look considerably bigger.”
Pause	After “we thought that looked rigged.”
Do not	Do not present day 0.8 vs day 42 as a field result. It is a simulation output . Say “in our model” every time.

Slide 9 — Architecture / Four layers

What it says	Sense, Interpret, Network, See — plus the ABC gotcha and the 500+ PACS context.
What it means	The ABC callout is the strongest “we have actually built this” detail in the deck.
Emphasise	“Every NDIR sensor auto-rezeros to the lowest value it sees. Inside grain it never meets fresh air — so that has to be disabled in firmware, or the sensor quietly erases the signal we exist to detect. That is the kind of thing you only find by building it.”
Vishal should deliver	the ABC point specifically. It is his domain and it lands better from the person who wrote the firmware.
Note	This slide says RH ± 2 %; slide 10 says ± 1.0 %. See §0.3 — datasheet band versus simulated sigma.

Slide 10 — Detection / We chose the explainable detector

What it says	Two rejected approaches, the chosen depth-matched differential, and the 1.5 / 3.0 / 4.2 sigma separation.
What it means	Showing your rejected options demonstrates engineering judgement far better than presenting only the winner.
60 seconds	“We did not use a fixed threshold, because godown temperature swings seasonally and any absolute limit either cries wolf in summer or sleeps through winter. We did not compare all nine ports together either, because depth stratification is real — about a thousand ppm — and pooling them buried the signal completely. What we do is compare each port only against the other lances at its own depth, on the same analyser. That cancels sensor drift and slow ventilation changes at the same time. Clean grain peaks at 1.5 sigma, we alert at 3.0, and a real hotspot reaches 4.2 — sustained over three consecutive readings, because one noisy sample is never an alarm.”
Judge will ask	“Where is the machine learning?” — See §28. The answer is that there is none, deliberately, and why.

Slide 11 — What We Built / State the boundary before they find it

What it says	A YES/NO table of what is and is not demonstrated, plus the digital twin’s capabilities.
What it means	The most honest slide in the deck, and therefore the most persuasive. The two NO rows are worth more than the four YES rows.
60 seconds	“Here is exactly what we have and have not shown. We have two sealed maize jars, one at storage moisture and one deliberately wetted. We have an SHT3x reading the headspace on the same ESP32 firmware built for the field probe. That reading goes into the same decision-engine code as the simulation — one source of truth. The dry jar comes back below the 0.82 growth floor; the wet jar above it, with a predicted onset. What we have <i>not</i> done: we have not watched grain spoil — they were sealed fifteen hours and fungus needs days. And we have not measured CO ₂ , because we could not source the sensor in time. Being precise about that boundary is the difference between a team that gets believed and one that gets caught.”
Pause	After “one that gets caught.”
Do not	Do not soften the NO rows. Their bluntness is the point.

Slide 12 — Deployment / Three lances, nine eyes

What it says	Lance geometry (0.8 / 2.0 / 3.2 m across; 0.5 / 1.2 / 2.0 m deep), the shared analyser, the I ² C design note, and the costed hardware.
Devavrath should deliver	the geometry and non-destructive insertion points.

Emphasise	“One analyser serves nine sampling points. That is a hardware decision, not a software one — and it does two things: it makes depth-resolved gas mapping affordable, and it removes inter-sensor drift, because every depth comparison is made by the same instrument against the same baseline.”
Judge may ask	“Why 1.2 m in the middle?” — it is the point furthest from any face and therefore naturally reads highest CO ₂ . The depth-matched comparison cancels exactly that.
Fix first	The “COVERAGE RESULT — to be inserted” placeholder must go. Either fill in the number or delete the box. A visible placeholder reads as unfinished work. Also see Finding 4 on the SHT31 price line.

Slide 13 — Intervention / The best treatment is the one you never need

What it says	Three escalating tiers — aerate, restack, targeted fumigation — plus verification.
What it means	This slide answers “so what?”. Detection without a response plan is just information. It also connects back to the resistance problem on slide 7.
60 seconds	“Early detection changes which tier you need. Tier one is automatic — drive existing godown aeration over the flagged zone. Reversible, non-toxic, cheap, and the system watches the CO ₂ and water activity fall to confirm it worked. Tier two is human hours: pull the affected bags, dry and separate them. That option only exists because you know <i>which</i> bags. Tier three is a licensed operator fumigating that stack, not the whole shed — less chemical, less exposure, less selection pressure for resistance. Late detection leaves you only tier three, applied to everything, on a chemical that resistance is already eroding.”
Emphasise	The verification step — “and nobody else does this.” If the anomaly does not fall, the system escalates rather than assuming success.

Slide 14 — Thank You

What to say	Do not end on “thank you”. End on the roadmap sentence, then invite questions:
Closing line	“The sensing hardware is commodity. The intelligence is the hard part, and that is what we built. Hardware validation and the mycology programme are the next phase, scoped and budgeted at ₹19,800 in our submitted proposal. Happy to take questions.”
Then	Stop talking. Let them ask.

37 Judge questions — the full bank

Seventy-five questions across eighteen categories. Format: **Short** is what you say first. **Technical** is where you go if pressed. **Never** is the trap to avoid.

Problem and context

Q1 Is this actually a big problem, or a student exercise?

- SHORT** India’s post-harvest cereal losses were assessed at 3.89 to 5.92 % in a NABCONS study reported to Parliament in 2024.
- TECHNICAL** Against production measured in hundreds of millions of tonnes, even the lower bound is millions of tonnes annually. And the exposure is growing — 500+ PACS godowns under construction, none instrumented.
- NEVER** Claim the entire loss figure is storage-stage. It is total post-harvest across all stages.

Q2 Why bag stacks rather than silos?

- SHORT** Because that is what India actually uses for public storage.
- TECHNICAL** Bag stacks have different access constraints (no auger, probing damages sacks), different air paths (voids between bags as well as kernels), and are built and dismantled rather than permanent — so permanent instrumentation is uneconomic. Commercial monitoring is built for the silo case.

Q3 Why is the interior invisible? Can't inspectors just look harder?

SHORT It is a reach problem, not a diligence problem.

TECHNICAL A spear reaches roughly 0.4 m; stacks are metres deep. FAO guidance also notes problems cluster non-uniformly and that absence of adult insects does not indicate absence of infestation, because eggs and larvae develop inside kernels.

NEVER Imply inspectors are careless. Frame it as geometry.

Q4 Why not just inspect more frequently?

SHORT Frequency does not fix reach. Inspecting twice as often still cannot reach 1.5 m in.

TECHNICAL Doubling inspection frequency roughly halves detection latency for events in the reachable band. For an interior pocket it changes nothing, because the pocket is never sampled at any frequency. Also, more inspection means more labour and more bag damage.

Q5 Is aflatoxin really the concern, or is this about weight loss?

SHORT Both, but aflatoxin is the one that is irreversible.

TECHNICAL Dry matter loss is recoverable in the sense that you lose a quantity and move on. Aflatoxin decides the fate of the consignment — it is not destroyed by drying or cooking, and it is a liver carcinogen. That asymmetry is why the problem is predictive rather than diagnostic.

Q6 Who is your actual customer?

SHORT PACS-level cooperative godowns first, then state warehousing and FCI.

TECHNICAL PACS facilities are the sharpest fit: new, numerous, and with no instrumentation or on-site storage scientists. They are also where per-facility cost sensitivity is highest, which is why the shared-analyser architecture matters commercially.

Grain science and biology

Q7 What is water activity, in one sentence?

SHORT The fraction of the water in the grain that is chemically free and therefore usable by an organism.

TECHNICAL Some water is bound to starch and protein and unavailable. Water activity is the free fraction on a 0–1 scale, and it equals the equilibrium relative humidity of the surrounding air. That equivalence is what lets a humidity sensor measure the biologically governing variable.

Q8 Why not just measure moisture content?

SHORT Because fungi respond to available water, not total water.

TECHNICAL Two batches at identical moisture content behave differently at different temperatures, because temperature changes how tightly the matrix holds water. Moisture content also requires a physical sample, so it cannot be continuous or interior.

Q9 Where does 0.82 come from?

SHORT It is the conservative end of the published range for *A. flavus* growth.

TECHNICAL Reported minima vary with strain, substrate and temperature. We deliberately took the lowest credible value, because for a warning system you want to alert early rather than late.

NEVER Say “below 0.82 it physically cannot grow”. See Finding 1.

Q10 Why is the toxin threshold higher than the growth threshold?

SHORT Toxin production is metabolically more demanding than growth alone.

TECHNICAL The organism can survive and grow in conditions where it does not produce aflatoxin. This is why we model two thresholds — growth at 0.82, toxin at 0.85 — rather than one.

Q11 What exactly is respiring — the grain or the fungus?

SHORT Both, and we cannot distinguish them from CO₂ alone.

TECHNICAL Grain is a living seed and respires at a low baseline rate. Above the water-activity threshold, fungal respiration dominates and rises steeply. Insects respire too. CO₂ integrates all three, which is a strength for detection and a limitation for diagnosis — and it is why water activity is measured alongside.

Q12 Why maize specifically?

SHORT Published isotherm coefficients exist for shelled corn, and it is the canonical aflatoxin substrate.

TECHNICAL Using ASAE D245 corn coefficients means our humidity-to-water- activity conversion is a published standard, not something we invented. It is also what is in our jars — one commodity validated properly rather than four claimed loosely.

Q13 Would this work for wheat or rice?

SHORT Yes, but it needs that commodity's isotherm coefficients.

TECHNICAL The physics and the detector are commodity-independent. Only the humidity-to-water-activity conversion is commodity-specific, and it is parameterised, so a new commodity is a configuration change rather than a redesign. The system should refuse to report a_w for an unconfigured commodity rather than silently using the wrong curve.

Q14 Your respiration coefficients — where are they from?

SHORT The functional form is standard; the coefficients are ours, tuned to reproduce published safe-storage timelines.

TECHNICAL Exponential in moisture and Q₁₀ in temperature is the standard form. Our R₀ = 0.5 mg CO₂/kg/day at 14 % and 25 °C, k = 0.7, Q₁₀ = 2.5. These are an engineering estimate, on Kritika's sign-off sheet.

NEVER Present them as measured values.

Q15 Is self-heating real, or theoretical?

SHORT Real, and well documented in stored-grain practice.

TECHNICAL It is a positive feedback: moisture raises respiration, respiration releases heat and water, which raise respiration further. In severe cases stacks heat substantially. It is also why spoilage is non-linear and why early detection is worth disproportionately more than late detection.

CO₂ and the physics

Q16 Why CO₂ rather than temperature?

SHORT Because it is the same event through a faster channel.

TECHNICAL CO₂ and heat come from one reaction, but CO₂ travels through the connected interstitial air while heat must conduct through the kernels, and grain is an insulator. Temperature is also an accumulated consequence — heat must build faster than it dissipates before a probe registers anything.

Q17 How do you know CO₂ arrives first?

SHORT From the diffusivities — and we have not yet measured it ourselves.

TECHNICAL Gas diffusivity in interstitial air is around 10^{-6} m²/s; bulk grain thermal diffusivity around 10^{-7} . Both scale as $t = L^2/D$, so the ratio is about 17 at any distance. CO₂ is also identified in storage-technology literature as an earlier indicator than temperature.

OUR SPECIFIC LEAD-TIME FIGURE IS MODELLED, NOT MEASURED, AND DEMONSTRATING IT IS THE POINT OF OUR VALIDATION EXPERIMENT.

NEVER Say “we measured a fifteen-times lead”.

Q18 Where does the 15× figure come from?

SHORT The ratio of the two diffusivities. It is 16.7; we round it down.

TECHNICAL 15.43 days for heat over 0.4 m divided by 0.93 days for CO₂ gives 16.7. Because both times scale with length squared, the distance cancels — the ratio holds at any separation, which is what makes it a robust claim even though the absolute times depend on assumed geometry.

Q19 Is comparing gas diffusivity with thermal diffusivity legitimate?

SHORT Yes — both have units of m²/s and obey the same scaling law.

TECHNICAL Thermal diffusivity $\alpha = k/(\rho c_p)$ has exactly the same dimensions as mass diffusivity D . Both describe spreading, both give $t = L^2/D$. It is a mathematically valid comparison, not an analogy.

Q20 Explain Fick’s law.

SHORT Gas flows from where there is a lot to where there is little, faster when the difference is steeper.

TECHNICAL $J = -D \, dC/dx$. Flux is proportional to the concentration gradient, negative because flow is down-gradient. The practical consequence is $t = L^2/D$ — distance squared, which is why interior placement matters so much.

Q21 Explain Fourier’s law.

SHORT Heat flows down a temperature gradient, at a rate set by the material.

TECHNICAL $q = -k \, dT/dx$. Grain has low thermal conductivity, and the trapped air that helps gas move actually hinders conduction — the same porosity works in opposite directions for the two signals.

Q22 Why is moisture so much slower than gas?

SHORT Because every kernel it passes absorbs some of it.

TECHNICAL Water vapour diffuses through air at a rate comparable to CO₂, but grain is a powerful moisture buffer. Vapour is repeatedly sorbed and desorbed, which enormously retards net transport. Our effective coefficient lumps that together and is the weakest of our three transport numbers.

Q23 What if the godown is ventilated?

SHORT It affects all ports similarly, and our comparison is differential.

TECHNICAL We never use an absolute threshold. Each port is compared against other lances at the same depth on the same analyser, so a facility-wide ventilation change is common-mode and cancels. A single anomalous port survives that cancellation; a ventilation change does not.

Q24 What if ambient CO₂ drifts?

SHORT Same answer — differential, plus a rolling per-port baseline.

TECHNICAL Seasonal or diurnal ambient change moves every port together. Only a local source produces port-to-port divergence at a given depth.

Q25 How do you know the CO₂ is from spoilage and not something else?

SHORT We do not, from CO₂ alone — which is why we measure water activity too.

TECHNICAL CO₂ localises and flags. Water activity tells us whether fungal growth is even possible at that location. If CO₂ is elevated and a_w is below the growth floor, that is a respiration anomaly without a fungal explanation — possibly insects, possibly grain respiration — and our action card says so rather than claiming mould.

Sensors

Q26 Why NDIR rather than a cheaper sensor?

SHORT Selectivity and long-term stability.

TECHNICAL Electrochemical cells consume reagent and have finite life; metal-oxide sensors are not CO₂-specific at all and drift heavily. Buried in grain you cannot recalibrate frequently, so drift and consumables are disqualifying. NDIR measures a real optical absorption specific to CO₂.

Q27 What is ABC and why does it matter?

SHORT Automatic Baseline Correction — and it would silently erase our signal.

TECHNICAL NDIR sensors assume they periodically see fresh 400 ppm air and re-zero to the lowest recent value. Buried in grain they never do, so ABC would progressively re-zero elevated interstitial CO₂ to “normal” while still reporting confident numbers. It must be disabled in firmware. This is the single most dangerous default in the whole system.

Q28 What happens if the sensor gets wet?

SHORT Hydrophobic membrane, plus the on-chip heater to clear condensation.

TECHNICAL Condensation pinning RH near 100 % is the most likely field failure. The SHT3x has an integrated heater that can be pulsed to drive off condensate, and we reject readings taken during a transient. Electronics are potted and isolated from the gas path.

Q29 How do you stop dust blocking the port?

SHORT Sintered filter plus several small perforations rather than one hole.

TECHNICAL Multiple small ports resist single-kernel blockage; a sintered element and PTFE membrane pass gas but stop fines. Firmware also watches for a port that goes flat while its peers vary, which is the signature of a blockage rather than a genuine reading.

Q30 ± 2 % RH is $\pm 0.02 a_w$. Your thresholds are 0.03 apart. Is that good enough?

SHORT It is tight, and we treat 0.82 as a conservative trigger rather than a precise boundary.

TECHNICAL This is a genuine limitation and worth stating. We mitigate three ways: take the conservative end of the threshold range; require the CO₂ anomaly to agree before escalating; and calibrate against saturated salt standards that bracket the decision region. A higher-grade sensor (± 1.5 %) is a drop-in upgrade if precision becomes limiting.

NEVER Claim the measurement is more precise than the sensor allows.

Q31 How will you calibrate in the field?

SHORT Saturated salt solutions for humidity; a reference gas or fresh-air check for CO₂.

TECHNICAL Saturated NaCl holds 0.753 a_w, KCl 0.843, K₂SO₄ 0.973 — a two-point calibration bracketing our decision region, using reagents any lab has. For CO₂, a manual fresh-air reference with ABC disabled, performed on a schedule rather than automatically.

Q32 Why one analyser rather than nine sensors?

SHORT Cost, and more importantly drift cancellation.

TECHNICAL Nine sensors would each drift independently, so port-to-port comparison — the basis of our whole detector — would become unreliable. One instrument reading all nine means every comparison is against the same baseline by the same hardware. It also removes the most fragile component from the buried environment.

ESP32 and electronics

Q33 Why ESP32 and not a Raspberry Pi?

SHORT Reliability and power. A Pi runs an OS that can corrupt on power loss.

TECHNICAL A sensing node must survive unattended power interruption. A microcontroller resumes; an OS may not. The ESP32 also deep-sleeps at sub-milliamp, has integrated Wi-Fi and BLE, two I²C controllers, and 3.3 V logic matching all our sensors — at around ₹500.

Q34 Why not an Arduino Uno?

SHORT No networking and 2 KB of RAM.

TECHNICAL We buffer readings when offline and compute an anomaly score locally. Neither is practical on an 8-bit part with 2 KB RAM and no radio.

Q35 How do three sensors share one I²C bus if they have two addresses?

SHORT A TCA9548A multiplexer — or the ESP32's second I²C controller.

TECHNICAL The SHT3x has only 0x44 and 0x45. Three on one bus collide. The multiplexer gives eight isolated channels and scales to nine sensors; alternatively the two hardware I²C controllers give two independent buses. Knowing both routes matters — the multiplexer is the scalable answer.

Q36 I²C over two metres is out of spec. How do you handle that?

SHORT Lower the clock, strengthen the pull-ups, use shielded cable.

TECHNICAL Drop to 100 kHz, use 2.2 kΩ pull-ups instead of 4.7 k, and shielded twisted pair. If it remains unreliable at full lance length the correct fix is a small microcontroller at each depth talking RS-485 — but we would measure the problem before designing for it.

Q37 What if there is no network?

SHORT The system is offline-first by design.

TECHNICAL The node computes its own anomaly score and can raise a local alarm with no connectivity at all. Readings buffer to flash and synchronise opportunistically. Rural godown connectivity cannot be assumed, so the decision logic never depends on a network.

Q38 How is it powered?

SHORT Mains where available, with deep-sleep duty cycling; solar is the field option.

TECHNICAL We sample hourly, not continuously, so the node sleeps at sub-milliamp between cycles. The supply must be sized for the simultaneous peak of Wi-Fi transmission and the analyser, not the average, or the node browns out during upload.

Mechanical and probe

Q39 How do you insert a 2 m probe without damaging bags?

SHORT Between bag courses, with a conical tip that parts grain rather than coring it.

TECHNICAL 20–25 mm diameter is a deliberate compromise — thin enough to pass between courses, thick enough to resist buckling under insertion load. The lance is removable and reusable across stacks, which matters because stacks are rebuilt.

Q40 How do you stop the three depth ports mixing inside the tube?

SHORT Internal seals between depth zones.

TECHNICAL Without them, all three ports share one internal volume and every port reads the average — destroying the depth resolution the whole detector depends on. This is the single most important internal detail of the lance.

Q41 What about dead volume in the tubing?

SHORT We purge each port before dwelling on it.

TECHNICAL Otherwise you measure the previous port's gas. Purge time is set by tube volume and pump rate, and is exactly the kind of question a CFD chamber-flow simulation answers with a real number rather than a guess.

Q42 Is CFD actually useful here or is it decoration?

SHORT Useful for two specific questions; decoration for anything larger.

TECHNICAL Chamber purge time and CO₂ transport from a pocket through porous grain to a port are both real design questions with real answers. A full 3D transient simulation of an entire godown would be dominated by porosity and boundary assumptions we cannot verify — a colourful picture and no added confidence.

NEVER Claim CFD proves the system works. It validates a model, not reality.

Q43 What material, and why?

SHORT Stainless 304 for production; aluminium for the prototype.

TECHNICAL Corrosion resistance in humid grain, food-contact acceptability, and stiffness for insertion. Aluminium is cheaper and lighter for a first article where we are testing geometry rather than durability.

Software, detection and AI

Q44 Where is the AI?

SHORT There is none, deliberately.

TECHNICAL We have nine sensors and no labelled failure dataset. Training a model on that would produce something we could neither explain nor defend. Every alert carries its own reasoning — this port, this many sigma above its depth peers, this water activity, this many days. When the validation programme produces a labelled dataset, machine learning becomes worth revisiting.

NEVER Call the risk score “AI”. It is a weighted sum, and saying so is stronger.

Q45 Why not a simple threshold?

SHORT Because there is no correct absolute value.

TECHNICAL Godown conditions swing seasonally, and so does baseline respiration. Any fixed limit either cries wolf in summer or sleeps through winter. We compare each port against its own depth peers instead.

Q46 Why compare within a depth rather than across all nine ports?

SHORT Depth stratification is real — about 1,000 ppm.

TECHNICAL Deep grain is furthest from a venting face and naturally reads highest. Pooling all nine ports buries a genuine local anomaly inside that natural spread. We found this in simulation: comparing all nine, the signal disappeared entirely.

Q47 What is your false-positive rate?

SHORT In simulation, clean grain peaks at 1.5 sigma against a 3.0 sigma threshold — but we have no field data.

TECHNICAL The persistence rule (three consecutive readings) suppresses single-sample noise. A real false-positive rate requires field deployment across seasons, and we would not claim one before we have it.

NEVER Quote a false-positive percentage. You do not have one.

Q48 Why simulate sensor noise at all?

SHORT So the detector is proven on imperfect data rather than idealised data.

TECHNICAL We inject datasheet-accurate noise — CO₂ ± 25 ppm plus 1.5 % of reading, RH ± 1 %, temperature ± 0.15 °C — into every virtual reading. A detector tuned on clean data would fail on first contact with a real sensor.

Q49 How do you fuse the signals?

SHORT A weighted composite — 35 % water activity, 40 % urgency, 25 % CO₂ anomaly.

TECHNICAL Each term is inspectable, so when an alert fires we can say precisely which component drove it. The weights are an assumption, not a derived result, and are on the verification sheet.

Q50 What if two signals disagree?

SHORT That is informative, and the action card says so.

TECHNICAL Elevated CO₂ with a_w below the growth floor means respiration without a fungal explanation — possibly insects, possibly a nearby source the port cannot resolve. The recommendation becomes “targeted probe” rather than “aerate for mould”. Disagreement changes the action rather than being suppressed.

Validation and evidence

Q51 What have you actually demonstrated?

SHORT A measured water-activity difference through real hardware and real code, plus a working simulation.

TECHNICAL Two sealed jars, SHT3x on the field firmware, readings into the same decision engine the simulation uses. Dry jar below the growth floor, wet jar above with a predicted onset. We have not observed spoilage and we have not measured CO₂.

Q52 So none of this is real?

SHORT The physics is simulated; the decision engine is production code; the water activity was measured.

TECHNICAL Be precise about which layer is which. The transport model is simulation. The interpretation pipeline is the same code that ran on our jar reading. The hardware exists and works. What is missing is the CO₂ sensor and field time.

Q53 How would you prove the CO₂ lead time?

SHORT Two instrumented grain columns, one with an induced wet pocket, logged continuously.

TECHNICAL CO₂, RH and temperature at multiple depths, sampled every 15–60 minutes for two to four weeks, against an identical control column. Acceptance criterion: CO₂ departs from control measurably earlier than temperature. That experiment is designed and not yet run — it needs the sensor.

Q54 Your jars were sealed 15 hours. Isn't that meaningless?

SHORT For spoilage, yes. For water activity, no.

TECHNICAL Sorption equilibrium between grain and headspace air happens in hours, so the a_w measurement is valid. Fungal growth takes days, so we make no claim about spoilage. That is exactly why slide 11 separates the two.

Q55 Is your baseline comparison fair?

SHORT We deliberately made it generous, and we can show you how.

TECHNICAL Fortnightly inspection when many facilities are monthly. Inspectors never miss visible mould. And we added a musty-smell detection route specifically because our first model had conventional inspection never finding the pocket, which looked rigged. Day 42 exists *because* we handicapped ourselves.

Q56 Where does 0.4 m spear reach come from?

SHORT It is our estimate of a bag spear's practical reach, and it needs a citable source.

TECHNICAL This is an assumption on the verification sheet. If it is too generous our comparison looks better than it should; if too harsh, worse. We believe it is realistic, and confirming it against a standard or a physical measurement is on our list.

NEVER Present it as a published figure.

Cost, scale and commercial

Q57 What does one unit cost?

SHORT Around ₹8,000–10,500 for the prototype, inside our ₹19,800 budget.

TECHNICAL The NDIR analyser dominates at around ₹5,000. That is exactly why one analyser serves nine ports rather than nine analysers serving nine ports.

Q58 Can this scale economically?

SHORT The expensive component is shared, so cost per monitored point falls with port count.

TECHNICAL Adding ports costs tubing and a manifold position, not another analyser. Our scale estimates are extrapolations from component pricing, not quotations — treat them as indicative.

Q59 Who pays for this?

SHORT The facility, justified by avoided loss and reduced fumigation.

TECHNICAL The honest answer is that a business case requires a pilot producing documented lead-time and chemical-use figures against that facility's existing practice. We have designed that pilot; we have not run it.

Q60 Why can't a commercial system do this?

SHORT They can — for silos, at silo prices.

TECHNICAL Commercial grain monitoring is capable and well engineered. Our claim is not that they cannot work; it is that they are designed for bulk vessels and priced for large operations, and Indian public storage is bagged stacks at PACS scale.

NEVER Claim competitors are technically incapable.

Deployment and practicality

Q61 How many lances per stack?

SHORT Three across a 4 m face, three depths each.

TECHNICAL That spacing is a design choice awaiting the coverage analysis — the study that determines what percentage of hotspots three lances at that spacing detect, and how quickly. It is on our list and not yet complete.

Q62 What about a pocket between lances?

SHORT That is a genuine coverage limitation.

TECHNICAL Diffusive reach determines the detection radius around each port. A pocket far enough from every port produces too small a differential. That is exactly what the coverage analysis quantifies, and the answer determines lance spacing.

Q63 Who installs and maintains it?

SHORT Insertion is designed to be repeatable by an unskilled operator.

TECHNICAL The lance is removable and reusable. Maintenance is filter replacement and periodic calibration. The analyser lives in an accessible head unit rather than buried, specifically so it can be serviced.

Q64 What if the operator ignores the alert?

SHORT Then nothing improves — which is why the output is an action, not a number.

TECHNICAL An action card naming stack, depth, action and deadline is far more actionable than a ppm reading. The tamper-evident log also creates accountability: what was flagged, when, and what was done.

Q65 Does it work during fumigation?

SHORT That is when verification matters most.

TECHNICAL Post-treatment CO₂ decay is our efficacy signal. If the anomaly does not fall after treatment, the system escalates rather than assuming success — which is the failure mode current practice never detects.

Safety, ethics and limitations

Q66 Are you culturing aflatoxin-producing fungi?

SHORT No. We use naturally occurring colonisation of wetted grain, in the lab, under supervision.

TECHNICAL No deliberate inoculation of toxigenic strains. Jars remain sealed outside the lab, handled with gloves and a dust mask, and autoclaved or disposed of per protocol. Deliberate culturing would require containment we do not claim to have.

Q67 What is the worst failure mode of your system?

SHORT A false negative that creates false confidence.

TECHNICAL A blocked port reading flat looks like healthy grain. That is worse than no system at all, because it displaces vigilance. We detect it by watching for ports that stop varying while peers vary — but it is the failure we take most seriously.

NEVER Claim the system cannot miss anything.

Q68 What are you most likely to be wrong about?

SHORT The absolute lead time. The ordering we are confident about; the magnitude is modelled.

TECHNICAL Real grain beds have convection, non-uniform porosity and bag boundaries our model treats simply. Those could compress or extend the gap. The physics that gas outruns heat is robust; the specific number of days is not.

Q69 What happens if your prediction is wrong?

SHORT The recommended first action is aeration — reversible, non-toxic and cheap.

TECHNICAL The tier structure is designed so a false positive costs very little: you aerate a zone that did not need it. Only confirmed, escalated cases reach fumigation. The asymmetry between the cost of a false alarm and the cost of a missed event is what justifies alerting early.

Novelty and positioning

Q70 What is genuinely novel here?

SHORT The application to bag stacks, the shared-analyser architecture, and prediction rather than measurement.

TECHNICAL The individual signals are established science and we say so. What is new is the combination applied to this geometry, the single-analyser design that makes depth-resolved mapping affordable while cancelling drift, and the shift from reporting a reading to issuing a dated recommendation with verification.

Q71 Isn't this just integration of existing parts?

SHORT Substantially yes — and the integration is the contribution.

TECHNICAL Conceding this is stronger than denying it. NDIR, isotherms and growth models all exist. Nobody has combined them for bagged Indian storage at a PACS-affordable price point with a closed verification loop. Naming honestly which parts are novel and which are integration is what makes the novel claims believable.

Q72 Why hasn't anyone done this?

SHORT Because the market that needs it cannot pay silo prices.

TECHNICAL The technology is not the barrier; the economics are. Commercial development targets large operations with capital budgets. The shared-analyser architecture is our attempt to make the same capability affordable at PACS scale.

Team

Q73 Why does this team need biotechnologists?

SHORT Because every threshold in the model is biological, and someone has to defend them.

TECHNICAL A team of only computer scientists could build the electronics and the dashboard but could not generate or defend the biological ground truth. Two biotechnology members give wet-laboratory validation capability a conventional software team cannot reproduce.

Q74 Who did what?

SHORT Three CSE on architecture, physics and firmware; two biotech on biology and baseline; one mechanical on the probe and transport.

TECHNICAL Name the specific ownership — Krittika owns the biological numbers, Netra the inspection baseline, Devavrath the transport constants and probe geometry. When a judge asks where a number came from, its owner answers.

Q75 What would you do with three more months?

SHORT Source the CO₂ sensor and run the column experiment.

TECHNICAL In priority order: demonstrate the CO₂ lead time in real grain; complete the coverage analysis to justify lance spacing; fabricate and insertion-test a real lance; and produce a labelled dataset that would make machine learning worth considering.

38 The twenty killer questions

These are the ones most likely to expose a weakness. If you know only twenty answers, know these.

#	QUESTION	THE ANSWER THAT SURVIVES
1	Why CO ₂ ?	Same event, faster channel. Gas moves through air gaps; heat conducts through insulating grain.
2	How do you know CO ₂ arrives before heat?	From published diffusivities — ratio ~17. And we have not measured it ourselves; that is what the validation experiment is for.
3	Where did 15× come from?	$15.43 \div 0.93 = 16.7$, rounded down. Distance cancels, so the ratio holds at any separation.
4	Why not temperature sensors?	They work — late. Heat arrives ~15× slower and attenuated. Also built for silos.
5	Why not moisture sensors?	We use them. But moisture only reports its own location and moves slowest of the three.
6	Why not thermal imaging?	Sees surfaces only. A 1.5 m deep pocket has no surface signature until very late.
7	Why not inspect more often?	Frequency does not fix reach. The pocket is never sampled at any frequency.
8	How do you know the CO ₂ is spoilage?	We do not, from CO ₂ alone. Water activity tells us whether growth is even possible there. That is why both are measured.
9	What if ambient CO ₂ changes?	Differential comparison, never absolute. Common-mode change cancels.
10	What if the godown is ventilated?	Same answer. It affects all ports at a depth equally.
11	What if the sensor gets wet?	Hydrophobic membrane plus the SHT3x on-chip heater. Condensation is our most likely field failure and we designed for it.
12	How deep can the probe really reach?	2.0–2.5 m by design. Not yet fabricated — that is design intent, not a demonstrated capability.
13	Why ESP32?	Integrated radio, deep sleep, dual I ² C, 3.3 V matching all sensors, survives power loss unlike an OS-based board, ~₹500.
14	How will you validate?	Two instrumented columns, one with an induced pocket, logged continuously for weeks. Designed, not yet run.

#	QUESTION	THE ANSWER THAT SURVIVES
15	What is novel?	Bag-stack geometry, shared-analyser drift cancellation, prediction over measurement, closed verification loop. The signals themselves are established.
16	What if the prediction is wrong?	First-tier action is aeration — reversible, non-toxic, cheap. False positives are designed to be inexpensive.
17	What does one unit cost?	~₹8,000–10,500, inside the ₹19,800 budget. The analyser dominates, which is why it is shared.
18	Can this scale?	Cost per monitored point falls with port count because the expensive part is shared. Scale figures are extrapolations, not quotes.
19	Where is the AI?	Deliberately absent. No labelled dataset, and an explainable detector we can defend beats a black box we cannot.
20	Is your baseline unfair?	We made it generous on purpose and added a detection route to help it. Day 42 exists because we handicapped ourselves.

Mathematics, prototype and reference

The equations you must be able to reproduce verbally, the demo plans, the glossary, and the emergency cheat sheet.

39 The mathematics, verbally defensible

39.1 Diffusion — the one equation to memorise

$$t = L^2 / D$$

Worked, for CO₂ over 0.4 m:

$$\begin{aligned} t &= (0.4)^2 / (2.0 \times 10^{-6}) \\ &= 0.16 / 0.000002 \\ &= 80,000 \text{ s} \\ &= 0.93 \text{ days} \end{aligned}$$

Same L, for heat ($\alpha = 1.2 \times 10^{-7}$):

$$\begin{aligned} t &= 0.16 / 0.00000012 \\ &= 1,333,333 \text{ s} \\ &= 15.4 \text{ days} \end{aligned}$$

$$\text{Ratio} = 15.4 / 0.93 = 16.7$$

If you can do this on a whiteboard, you own slide 3. The key point to add: because L is squared in both, it cancels in the ratio — so 16.7 holds at any distance, even though the absolute days depend on the geometry you assume.

39.2 Mass balance — CO₂ accumulation

$$\text{Rate of change} = \text{production} - \text{removal}$$

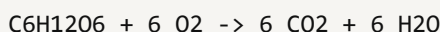
$$dC/dt = (R \times \rho_{\text{bulk}}) / (\text{porosity} \times \text{conversion}) - k_{\text{vent}} \times (C - C_{\text{ambient}})$$

At steady state, $dC/dt = 0$, so:

$$C_{\text{steady}} = C_{\text{ambient}} + \text{production} / k_{\text{vent}}$$

The useful intuition: **a pocket does not raise CO₂ without limit.** It rises until production balances loss to the venting faces. That equilibrium level, not an unbounded climb, is what the sensor sees.

39.3 Stoichiometry — converting CO₂ to grain lost



$$180 \text{ g glucose} \rightarrow 264 \text{ g CO}_2$$

$$\text{So } 1 \text{ g dry matter lost} \rightarrow 1.467 \text{ g CO}_2$$

$$1\% \text{ dry matter loss} \rightarrow 14.67 \text{ g CO}_2 \text{ per kg grain}$$

This is chemistry, not a model parameter. It lets accumulated CO₂ be converted directly into grain actually lost — which is how the system reports dry-matter loss.

39.4 The isotherm

```
aw = exp( -(A / (T + C)) x exp( -B x M_db ) )  
A = 620.7, B = 0.2118, C = 52.15 (shelled corn, ASAE D245)
```

```
Check: M = 14% wb -> 16.28% db, T = 25 C  
exp(-0.2118 x 16.28) = 0.0318  
620.7 / (25 + 52.15) = 8.045  
aw = exp(-8.045 x 0.0318) = 0.774 [matches slide 5]
```

39.5 Growth rate and days to onset

```
mu = mu_opt x f(aw) x f(T)  
  
f(aw) = (aw - 0.82) / (0.99 - 0.82), clamped 0..1  
f(T) = 1 - ((T - 33)/w)^2, clamped 0..1  
mu_opt = 0.20 /day
```

```
days_to_onset = 1 / mu
```

```
Check: aw 0.87, T 30 C  
f(aw) = 0.05/0.17 = 0.294  
f(T) = 1 - (3/21)^2 = 0.980  
mu = 0.20 x 0.294 x 0.980 = 0.0576  
days = 1/0.0576 = 17.4 [matches slide 5]
```

39.6 The anomaly score

```
For port i at depth d:  
peers = other ports at the same depth  
z_i = (reading_i - mean(peers)) / spread(peers)  
  
ALERT if z_i >= 3.0 for 3 consecutive readings
```

40 Prototype and demo plans

40.1 Minimum viable prototype

ELEMENT	SPECIFICATION	STATUS
CO ₂ sensor	NDIR, 0–40,000 ppm (SCD30 class), ABC disabled	Not sourced
Humidity/temp	3× SHT3x, I ² C, on-chip heater	Built
Controller	ESP32 DevKit, 3.3 V I ² C throughout	Built
Bus expansion	TCA9548A multiplexer	Have
Local display	SSD1306 OLED — standalone readout with no laptop	Built
Power	USB for bench; 5 V regulated for field	Built

ELEMENT	SPECIFICATION	STATUS
Probe	Perforated lance, 3 depths, inter-port seals	Design only
Software	Digital twin, decision engine, bench-reading panel	Built
Validation	Two sealed maize jars, control and wetted	Running

40.2 The five-minute demo

1. **0:00** Frame it — 4×2.5 m stack in cross-section, three lances, nine ports.
2. **0:30** Show all four overlays flat and boring. “Remember what this looks like.”
3. **1:00** Inject a hotspot — **ask the judge where.**
4. **1:30** CO₂ view: the plume blooms within a simulated day.
5. **2:00** Switch to temperature: almost nothing. **This is the moment.**
6. **2:30** The sigma chart — green CO₂ crossing the threshold, red temperature flat.
7. **3:00** The action card. Read the evidence lines aloud.
8. **3:30** Accelerate to day 60. Comparison timeline: day 0.8 versus day 42.
9. **4:30** The jars — type a real reading into the bench panel.

40.3 The ten-minute demo

The five-minute run, plus:

- **Run a second preset** (monsoon roof leak) to show reproducibility rather than a lucky run.
- **Show the self-check** passing in the header — the science verifies itself before every session.
- **Open slide 8** and walk the baseline assumptions, explaining that you weakened your own number.
- **Read both jars** — control and wetted — and let the engine return opposite verdicts.

40.4 The thirty-minute technical demo

Everything above, plus:

- **Open the source.** Walk the constants block: every value carries its source in a comment.
- **Show the detector** comparison — run with all-nine-port pooling to demonstrate the signal disappearing, then depth-matched to show it return. **This is the most convincing thing you can show a technical judge.**
- **Derive $t = L^2/D$** on paper and reproduce the 0.9 / 15.4 figures live.
- **Show the firmware** — CRC validation, ABC disabling, purge-and-dwell sequencing.
- **Walk the FMEA** — particularly the false-negative case, and why it is the one you take most seriously.
- **State the validation gap** and the experiment designed to close it.

41 Glossary

TERM	SIMPLE → TECHNICAL, AND THE SENTENCE TO SAY
Water activity (a_w)	The usable fraction of water, 0–1. Technically the ratio of vapour pressure over the material to that over pure water; equals equilibrium relative humidity divided by 100. “ <i>Moisture is the total; water activity is the spendable part.</i> ”
Moisture content	Total water as a percentage of mass. Wet basis divides by total mass, dry basis by dry matter. “ <i>It counts water the fungus cannot reach.</i> ”
Relative humidity	How saturated the air is. “ <i>At equilibrium, the humidity between the kernels is the grain’s water activity — that equivalence is the whole sensing strategy.</i> ”

TERM	SIMPLE → TECHNICAL, AND THE SENTENCE TO SAY
Interstitial air	The connected air in the gaps between kernels. <i>“About forty percent of a grain bed by volume, and the channel our gas signal travels through.”</i>
Porosity	The void fraction of the bed. <i>“The same porosity that lets gas through is what makes grain insulate — one property, opposite effects.”</i>
Respiration	Sugar plus oxygen to CO ₂ , water and heat. <i>“Three products of one reaction — which is why one can stand in for the others.”</i>
Diffusion	Spreading down a concentration gradient. <i>“Fick’s law; time goes as distance squared, which is why interior placement matters.”</i>
Thermal diffusivity	How fast temperature spreads, $\alpha = k/(\rho c_p)$. <i>“Same units as gas diffusivity, which is what makes the comparison legitimate.”</i>
Fick’s law	$J = -D \, dC/dx$. <i>“Gas flows from more to less, faster when the gradient is steeper.”</i>
Fourier’s law	$q = -k \, dT/dx$. <i>“Heat flows down a temperature gradient at a rate set by the material — and grain is an insulator.”</i>
Sorption isotherm	The curve linking moisture, temperature and water activity. <i>“We use modified Chung–Pfoest with published ASAE coefficients — a standard, not an invention.”</i>
NDIR	Non-dispersive infrared. <i>“CO₂ absorbs at 4.26 microns; we measure how much light survives the gas cell. Selective and stable for years.”</i>
ABC	Automatic Baseline Correction. <i>“The sensor re-zeros to the lowest value it sees. Buried in grain that would erase our signal, so we disable it.”</i>
Aspergillus flavus	A storage fungus that produces aflatoxin. <i>“Grows above roughly 0.82 water activity; produces toxin only in a narrower window.”</i>
Aflatoxin	A liver carcinogen, heat stable. <i>“You can kill the fungus. You cannot unmake the toxin — which is why the problem is predictive.”</i>
Mycotoxin	Any toxic fungal metabolite. Aflatoxin is one.
Dry matter loss	Grain substance consumed by respiration. <i>“One percent loss equals 14.67 grams of CO₂ per kilogram — so accumulated gas converts directly into grain lost.”</i>
Anomaly detection	Flagging what differs from its peers rather than from a fixed number. <i>“Because there is no correct absolute CO₂ value in a godown.”</i>
Sensor fusion	Combining weak signals into a stronger estimate. <i>“CO₂ says something is respiring; water activity says whether fungus is possible there. Together they decide.”</i>
ESP32	Microcontroller with integrated Wi-Fi and BLE. <i>“Survives power loss, deep-sleeps at microamps, and speaks 3.3 V I²C like all our sensors.”</i>
I²C	Two-wire bus where devices share lines by address. <i>“Which is why an address collision between three identical sensors is a real design problem.”</i>
MQTT	Lightweight publish/subscribe messaging for intermittent links. <i>“Suited to a node that is offline much of the time.”</i>
Edge computing	Processing on the device rather than in the cloud. <i>“Our node computes its own anomaly score, so it works with Wi-Fi off.”</i>
Cardinal model	Growth model defined by minimum, optimum and maximum. <i>“Standard predictive mycology — we did not invent the approach.”</i>
Phosphine	The dominant grain fumigant. <i>“Indian Tribolium populations show 2.11 to 70.94-fold resistance — blanket dosing selects for exactly that.”</i>

Anna-Drishti in one page

The problem

Indian grain sits in bagged stacks metres deep. Damp pockets form inside from roof leaks, wet consignments or condensation. The interior cannot be inspected — a spear reaches ~0.4 m. Spoilage becomes visible only after it is irreversible, and aflatoxin cannot be undone by drying or cooking. Losses assessed at **3.89–5.92 %** (NABCONS, to Parliament, 2024). 500+ PACS godowns under construction, none instrumented.

The insight

Spoilage is **respiration**: sugar + oxygen → CO₂ + water + heat, from one reaction. The three products travel at wildly different speeds. Over 0.4 m: **CO₂ 0.9 days, heat 15.4 days, moisture 123.5 days**. Gas leads by ~17×. We stand where the signal arrives first.

The science

Fungi respond to **water activity**, not moisture — the free, usable fraction. At equilibrium, interstitial RH is a_w . *A. flavus* growth floor **0.82** (conservative end of a published range); toxin needs **0.85**. Crossing point is about **15 % moisture**. Above it, the clock starts — and the clock is what we report.

The solution

Sense (9 ports, 3 lances × 3 depths, one shared NDIR analyser) → **Interpret** (RH → a_w → growth forecast) → **Decide** (composite risk, ranked ports, action card with deadline) → **Verify** (post-treatment signature decay). Output is a port, a reason and a deadline — not an alarm.

The hardware

SCD30-class NDIR (0–40,000 ppm, **ABC disabled**), 3× SHT3x, ESP32, TCA9548A multiplexer. All 3.3 V I²C by design. ~**₹8,000–10,500**, inside the ₹19,800 budget. Offline-first.

The detector

Not a fixed threshold (seasonal swings) and not all nine ports pooled (1,000 ppm depth stratification buries the signal). **Depth-matched differential** — each port against other lances at its own depth, same analyser, cancelling drift and ventilation together. Clean grain **1.5σ**, alert at **3.0σ**, hotspot **4.2σ**, sustained 3 readings.

The novelty

Signals are established science. New: bag-stack geometry, single-analyser drift cancellation, prediction rather than measurement, closed verification loop.

The validation

Have: two sealed maize jars, SHT3x on field firmware, reading into the same decision engine, dry jar below 0.82 and wet jar above with a predicted onset; a self-checking digital twin. **Do not have**: observed spoilage, any CO₂ measurement, field lead-time data, the coverage result.

Judge defence

Key numbers

0.9 / 15.4 / 123.5 d	CO ₂ / heat / moisture over 0.4 m	16.7×	true gas:heat ratio (deck says ~15)
0.82 / 0.85	growth floor / toxin line	14.66 %	moisture at the 0.82 crossing, 30 °C
1.5 / 3.0 / 4.2 σ	clean / alert / hotspot	0.79 vs 42 d	our detection vs conventional, <i>in simulation</i>
2.11–70.94×	phosphine resistance, <i>Sci Rep</i> 13:16497 (2023)	3.89–5.92 %	post-harvest loss, NABCONS 2024
14.67 g/kg	CO ₂ per 1 % dry matter lost	~₹8–10.5k	prototype cost

Key equations

$t = L^2 / D$	diffusion time (distance SQUARED)
$\alpha = k / (\rho \cdot c_p)$	thermal diffusivity - same units as D
$a_w = RH / 100$	at equilibrium
1% DML = 14.67 g CO ₂ /kg	
$z = (\text{port} - \text{mean}(\text{depth peers})) / \text{spread}$	

The twenty answers, compressed

Why CO ₂ ?	Same event, faster channel
How do you know?	Diffusivity ratio ~17. Not measured by us yet
Where is 15×	15.43÷0.93=16.7, rounded. Distance cancels
Why not temperature?	Works, but ~15× later and attenuated
Why not moisture?	We use it. Slowest signal; reports only its own spot
Why not thermal imaging?	Surfaces only; 1.5 m deep has no signature
Inspect more often?	Frequency does not fix reach
Is CO ₂ from spoilage?	Unknown alone — that is why a_w too
Ambient CO ₂ drift?	Differential, never absolute
Ventilation?	Common-mode, cancels
Sensor gets wet?	Membrane + on-chip heater. Most likely field failure
Probe depth?	2–2.5 m design intent. Not fabricated
Why ESP32?	Radio, deep sleep, dual I ² C, 3.3 V, survives power loss
Validation?	Two columns, induced pocket, weeks of logging. Not yet run
Novelty?	Geometry, shared analyser, prediction, verification
Prediction wrong?	Tier 1 is aeration — reversible and cheap
Cost?	~₹8–10.5k. Analyser dominates, hence shared
Scale?	Cost per point falls with port count
Where is AI?	None, deliberately. No labelled data; explainability wins
Baseline unfair?	Made generous on purpose; we added a route to help it

The four things never to say

1. **“A. flavus physically cannot grow below 0.82.”** Say: conservative end of a published range.
2. **“We measured a 15× lead time.”** Say: our model gives ~17× from published diffusivities; we have not measured it.
3. **“We watched grain spoil”** or **“we measured CO₂”**. Neither is true.
4. **“Existing systems can’t do this.”** They can — for silos, at silo prices. The argument is geometry and economics.

The three sentences that win the room

The insight: “The same biological event announces itself as gas about two weeks before it announces itself as warmth. We chose to stand where it arrives first.”

The honesty: “We do not measure toxin. We predict the conditions that create it, and we say so in every output the system produces.”

The fairness: “Our first model had conventional inspection never finding this pocket. We thought that looked rigged, so we added a detection route to help the baseline. Day 42 exists because we handicapped ourselves.”